

Kafka interview questions

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Give all Kafka interview questions and answers

Kafka Interview Questions & Answers Core Concepts 1. What is Apache Kafka? Apache Kafka is a distributed, fault-toleran...

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Kafka Day-to-Day Work Questions & Answers CLI & Operations 1. How do you create a Kafka topic from the command lin...

More scenario based q&a on Kafka

Kafka Scenario-Based Questions & Answers Data Loss & Durability Scenarios 1. Your team reports that some messages a...

Any topics related to Kafka we missed

Kafka Topics We Haven't Covered Yet Quick Overview of Missing Areas Here's a comprehensive map of what we haven't...

How to connect if Kafka is using certificates How to handle serialization on Kafka How to configure Kafka

Kafka Certificates, Serialization & Configuration PART 1: Connecting to Kafka with Certificates (SSL/TLS) How SSL Works in...

Did we cover most interview related to Kafka or anything pending

Kafka Interview Coverage Review What We've Covered  Area Topics Covered Core Concepts Topics, partitions, offsets,...

What is throughput here When rebalancing happens Kafka must be doing slow processing right

Great questions! Let me explain both clearly. 1. What is Throughput in Kafka? Throughput = how much data Kafka can...

Go deeper on these topics performance tuning, or system design with Kafka?

Kafka Performance Tuning & System Design — Deep Dive PART 1: Performance Tuning The Performance Tuning Mental...

Kafka Interview Questions & Answers

Core Concepts

1. What is Apache Kafka? Apache Kafka is a distributed, fault-tolerant, high-throughput publish-subscribe messaging system designed for real-time data streaming. It was originally developed at LinkedIn and later open-sourced under the Apache Foundation.

2. What are the key components of Kafka?

- **Producer** – publishes messages to topics
- **Consumer** – subscribes to topics and reads messages
- **Broker** – a Kafka server that stores and serves messages
- **Topic** – a category/feed to which messages are published
- **Partition** – topics are split into partitions for parallelism
- **Zookeeper / KRaft** – manages broker metadata and leader election
- **Consumer Group** – a group of consumers sharing partition load

3. What is a Kafka Topic and Partition? A topic is a logical channel for messages. Each topic is divided into partitions, which are ordered, immutable logs. Partitions enable parallelism — multiple consumers can read from different partitions simultaneously.

4. What is the role of an offset in Kafka? An offset is a unique sequential ID assigned to each message within a partition. Consumers track offsets to know which messages have been read. Offsets are per-partition and allow consumers to resume from where they left off.

Architecture

5. What is a Kafka Broker? A broker is a Kafka server that receives messages from producers, stores them on disk, and serves them to consumers. A Kafka cluster typically has multiple brokers for fault tolerance and scalability.

6. What is a Kafka Cluster? A Kafka cluster is a group of brokers working together. One broker acts as the **Controller** (manages partition leadership). Data is replicated across brokers for durability.

7. What is a Leader and Follower in Kafka? Each partition has one **Leader** broker that handles all reads and writes, and one or more **Follower** brokers that replicate the data. If the leader fails, a follower is elected as the new leader.

8. What is ISR (In-Sync Replicas)? ISR is the set of replicas that are fully caught up with the leader. Only ISR members are eligible for leader election. If a follower falls too far behind, it is removed from the ISR.

9. What is ZooKeeper's role in Kafka? Is it still needed? Traditionally, ZooKeeper managed broker metadata, leader election, and topic configs. Since Kafka 2.8+, **KRaft mode** (Kafka Raft) allows Kafka to run without ZooKeeper, using an internal metadata quorum instead. ZooKeeper is fully deprecated in newer versions.

Producers

10. How does a Kafka Producer work? A producer serializes a message, determines the target partition (via key hash, round-robin, or custom partitioner), and sends it to the leader broker of that partition. It can batch messages for efficiency.

11. What are the producer acknowledgment modes (acks)?

- `acks=0` – No acknowledgment; fastest but data can be lost
- `acks=1` – Leader acknowledges; moderate safety
- `acks=all` (or `-1`) – All ISR replicas acknowledge; strongest durability guarantee

12. What is idempotent producer in Kafka? Setting `enable.idempotence=true` ensures a producer doesn't write duplicate messages even on retries. Kafka assigns each producer a PID and tracks sequence numbers per partition.

13. What is a Kafka Transaction? Kafka transactions allow atomic writes across multiple partitions/topics. A transaction either fully commits or fully aborts — useful for exactly-once processing.

Consumers

14. What is a Consumer Group? A consumer group is a set of consumers that jointly consume a topic. Each partition is assigned to exactly one consumer within the group, enabling parallel consumption. Multiple groups can independently consume the same topic.

15. What is the difference between `earliest` and `latest` offset reset?

- `earliest` – Consumer reads from the very beginning of the topic
- `latest` – Consumer reads only new messages published after it started

16. How does Kafka handle Consumer Rebalancing? When consumers join or leave a group, Kafka triggers a **rebalance** — reassigning partitions among active consumers. During rebalance, consumption is paused. Strategies include Range, RoundRobin, and Sticky (minimizes partition movement).

17. What is the difference between `at-most-once`, `at-least-once`, and `exactly-once` delivery?

- **At-most-once** – messages may be lost, never duplicated (commit before processing)
 - **At-least-once** – messages may be duplicated, never lost (commit after processing)
 - **Exactly-once** – guaranteed no loss, no duplicates (requires idempotent producers + transactions)
-

Storage & Performance

18. How does Kafka store messages? Kafka stores messages as sequential log segments on disk per partition. Messages are immutable once written. Kafka uses zero-copy transfer (sendfile) for high-throughput reads.

19. What is Kafka's retention policy? Kafka retains messages based on:

- **Time** – e.g., keep messages for 7 days (`log.retention.hours`)
- **Size** – e.g., keep up to 1 GB per partition (`log.retention.bytes`)
- **Compaction** – keeps only the latest value per key (log compaction)

20. What is Log Compaction? Log compaction retains only the most recent message for each key in a topic. Useful for maintaining the latest state (like a changelog). Older duplicate-key records are eventually removed.

21. Why is Kafka so fast?

- Sequential disk writes (append-only log)
- Zero-copy data transfer
- Message batching and compression
- Partition-based parallelism
- OS page cache utilization

Advanced Topics

22. What is Kafka Streams? Kafka Streams is a lightweight Java library for building real-time stream processing applications on top of Kafka. It supports stateless (filter, map) and stateful (aggregations, joins, windowing) operations without needing a separate processing cluster.

23. What is KSQL / ksqlDB? ksqlDB is a SQL-like streaming database built on Kafka Streams. It allows you to write SQL queries over Kafka topics in real time — great for filtering, aggregations, and joins without writing Java code.

24. What is a Kafka Connect? Kafka Connect is a framework for streaming data between Kafka and external systems (databases, S3, Elasticsearch, etc.) using pre-built **connectors** — without writing custom code.

25. What is the difference between Kafka and RabbitMQ?

Feature	Kafka	RabbitMQ
Model	Log-based pub/sub	Message queue
Message Retention	Configurable (days)	Deleted after consumption
Throughput	Very high	Moderate
Ordering	Per partition	Per queue
Use Case	Streaming, event sourcing	Task queues, RPC

26. What is Schema Registry? Schema Registry (by Confluent) stores and enforces Avro/JSON/Protobuf schemas for Kafka messages. It ensures producers and consumers agree on message structure, preventing deserialization errors.

27. How do you secure a Kafka cluster?

- **Authentication** – SSL/TLS or SASL (PLAIN, SCRAM, Kerberos)
- **Authorization** – ACLs (Access Control Lists) via Kafka's built-in ACL manager
- **Encryption** – TLS for data in transit; OS-level encryption for data at rest

Scenario-Based Questions

28. How would you handle a slow consumer in Kafka?

- Increase consumer parallelism by adding more partitions and consumers

- Optimize consumer processing (batching, async processing)
- Monitor `consumer lag` using tools like Kafka UI or Burrow
- Scale horizontally with more consumer group members

29. How do you achieve exactly-once semantics end-to-end? Enable idempotent producer (`enable.idempotence=true`), use Kafka transactions (`transactional.id`), and set consumer `isolation.level=read_committed` to only read committed messages.

30. What happens when a broker goes down? Kafka detects the failure via heartbeat timeouts. The Controller triggers leader election for affected partitions — electing a new leader from the ISR. Producers and consumers automatically reconnect to the new leader with minimal disruption.

These cover beginner through senior-level Kafka interviews. Want me to go deeper on any specific area like Kafka Streams, performance tuning, or system design with Kafka?

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Kafka Day-to-Day Work Questions & Answers

CLI & Operations

1. How do you create a Kafka topic from the command line?

```
kafka-topics.sh --create \  
  --bootstrap-server localhost:9092 \  
  --topic my-topic \  
  --partitions 3 \  
  --replication-factor 2
```

- `--partitions` – controls parallelism
- `--replication-factor` – controls fault tolerance (should not exceed broker count)

2. How do you list all topics in a Kafka cluster?

```
kafka-topics.sh --list --bootstrap-server localhost:9092
```

3. How do you describe a topic (partitions, replicas, ISR)?

```
kafka-topics.sh --describe \  
  --bootstrap-server localhost:9092 \  
  --topic my-topic
```

Output shows Leader, Replicas, and ISR for each partition — useful for spotting under-replicated partitions.

4. How do you produce test messages from the terminal?

```
kafka-console-producer.sh \  
  --bootstrap-server localhost:9092 \  
  --topic my-topic
```

Type messages and hit Enter. To send with keys:

```
kafka-console-producer.sh \  
  --bootstrap-server localhost:9092 \  
  --topic my-topic \  
  --property "key.separator=:" \  
  --property "parse.key=true"
```

5. How do you consume messages from the terminal?

```
# Read new messages only
kafka-console-consumer.sh \
  --bootstrap-server localhost:9092 \
  --topic my-topic

# Read from the beginning
kafka-console-consumer.sh \
  --bootstrap-server localhost:9092 \
  --topic my-topic \
  --from-beginning

# Read with keys and timestamps
kafka-console-consumer.sh \
  --bootstrap-server localhost:9092 \
  --topic my-topic \
  --from-beginning \
  --property print.key=true \
  --property print.timestamp=true
```

Consumer Lag & Monitoring

6. How do you check consumer group lag?

```
kafka-consumer-groups.sh \
  --bootstrap-server localhost:9092 \
  --describe \
  --group my-consumer-group
```

Output shows **LOG-END-OFFSET**, **CURRENT-OFFSET**, and **LAG** per partition. High lag means consumers are falling behind producers.

7. What does consumer lag mean and how do you fix it? Lag = Log-End-Offset – Current-Offset. It means consumers haven't processed all available messages.

Fixes:

- Add more consumers (up to partition count)
- Increase **max.poll.records** to process more per poll
- Optimize consumer processing logic (async, batching)
- Increase partitions to allow more parallelism
- Check for GC pauses or slow downstream systems

8. How do you list all consumer groups?

```
kafka-consumer-groups.sh \
  --bootstrap-server localhost:9092 \
  --list
```

9. How do you reset consumer group offsets?

```
# Reset to earliest
kafka-consumer-groups.sh \
  --bootstrap-server localhost:9092 \
  --group my-consumer-group \
  --topic my-topic \
  --reset-offsets \
  --to-earliest \
  --execute

# Reset to a specific datetime
kafka-consumer-groups.sh \
  --bootstrap-server localhost:9092 \
  --group my-consumer-group \
  --topic my-topic \
  --reset-offsets \
  --to-datetime 2024-01-15T00:00:00.000 \
  --execute

# Reset to latest
kafka-consumer-groups.sh \
  --bootstrap-server localhost:9092 \
  --group my-consumer-group \
  --topic my-topic \
  --reset-offsets \
  --to-latest \
  --execute
```

⚠ Consumer group must be **inactive** (stopped) before resetting offsets.

Topic Management

10. How do you increase partition count for a topic?

```
kafka-topics.sh --alter \
  --bootstrap-server localhost:9092 \
  --topic my-topic \
  --partitions 6
```

⚠ You can only **increase** partitions, never decrease. Increasing partitions changes key-based routing for new messages.

11. How do you delete a Kafka topic?

```
kafka-topics.sh --delete \
  --bootstrap-server localhost:9092 \
  --topic my-topic
```

Requires `delete.topic.enable=true` in broker config.

12. How do you change topic-level configs (e.g., retention)?

```
# Set retention to 2 days
kafka-configs.sh --alter \
  --bootstrap-server localhost:9092 \
  --entity-type topics \
  --entity-name my-topic \
  --add-config retention.ms=172800000

# Describe current topic configs
kafka-configs.sh --describe \
  --bootstrap-server localhost:9092 \
  --entity-type topics \
  --entity-name my-topic
```

13. How do you check under-replicated partitions?

```
kafka-topics.sh \
  --bootstrap-server localhost:9092 \
  --describe \
  --under-replicated-partitions
```

Under-replicated partitions indicate a broker is down or a follower is lagging — needs immediate attention.

Producer Configurations (Day-to-Day Tuning)

14. What producer configs do you tune most often in practice?

Config	Purpose	Common Value
acks	Durability guarantee	all for production
retries	Auto-retry on failure	Integer.MAX_VALUE
retry.backoff.ms	Wait between retries	100-500ms
linger.ms	Batch wait time	5-20ms for throughput
batch.size	Max batch bytes	16384-65536
compression.type	Message compression	snappy or lz4
enable.idempotence	No duplicate writes	true
max.block.ms	Max wait if buffer full	60000

15. What happens when the producer buffer is full? If buffer.memory is exhausted, the producer blocks for max.block.ms. If still full after that, it throws a BufferExhaustedException. Fix by increasing buffer.memory, reducing linger.ms, or scaling up producers.

Consumer Configurations (Day-to-Day Tuning)

16. What consumer configs do you tune most often?

Config	Purpose	Common Value
<code>max.poll.records</code>	Records per poll	<code>500</code>
<code>max.poll.interval.ms</code>	Max processing time per poll	<code>300000</code>
<code>session.timeout.ms</code>	Heartbeat timeout	<code>45000</code>
<code>heartbeat.interval.ms</code>	Heartbeat frequency	<code>3000</code>
<code>fetch.min.bytes</code>	Min data per fetch	<code>1-50000</code>
<code>fetch.max.wait.ms</code>	Max wait for min bytes	<code>500</code>
<code>auto.offset.reset</code>	Behavior on no offset	<code>earliest</code> or <code>latest</code>
<code>enable.auto.commit</code>	Auto commit offsets	<code>false</code> (manual in prod)

17. Why would you set `enable.auto.commit=false` in production? Auto-commit commits offsets at a fixed interval regardless of whether messages were successfully processed. If your app crashes between commit and processing, messages are lost. Manual commit (`commitSync` / `commitAsync`) ensures you only commit after successful processing.

Debugging Common Issues

18. How do you debug a consumer that keeps rebalancing? Frequent rebalancing is usually caused by:

- Consumer taking too long to process → increase `max.poll.interval.ms`
- Consumer not sending heartbeats → increase `session.timeout.ms` or reduce processing time
- Consumers crashing due to exceptions → check application logs
- Deploying new instances frequently → use **Static Group Membership** (`group.instance.id`) to avoid rebalance on restart

19. How do you handle duplicate message processing?

- Make consumer logic **idempotent** (safe to process same message twice)
- Use a deduplication store (Redis, DB) keyed on message offset or a business ID
- Enable exactly-once with `enable.idempotence=true` + transactions on producer side
- Track processed offsets in your own DB and skip already-seen messages

20. What do you do when a topic has messages stuck (not consumed)?

```
# Step 1: Check lag
kafka-consumer-groups.sh --bootstrap-server localhost:9092 \
  --describe --group my-group

# Step 2: Check if consumers are active
# Look for "CONSUMER-ID" column - empty means no active consumers

# Step 3: Read a sample message to inspect
kafka-console-consumer.sh \
```

```
--bootstrap-server localhost:9092 \
--topic my-topic --from-beginning --max-messages 5

# Step 4: Reset offset if needed (stop consumers first)
kafka-consumer-groups.sh --bootstrap-server localhost:9092 \
--group my-group --topic my-topic \
--reset-offsets --to-earliest --execute
```

21. How do you find which broker is the leader for a partition?

```
kafka-topics.sh --describe \
--bootstrap-server localhost:9092 \
--topic my-topic
```

The **Leader** column shows the broker ID. Cross-reference with broker configs to find the host.

22. How do you handle a poison pill message (message causing consumer to crash)? A poison pill is a malformed/unexpected message that crashes your consumer repeatedly.

Strategies:

- Wrap processing in try-catch, log the bad message, and skip it
- Route failed messages to a **Dead Letter Topic (DLT)**
- Use Spring Kafka's `@RetryableTopic` or a `DeadLetterPublishingRecoverer`
- Alert on DLT message count as a health signal

```
// Example: Dead Letter Topic pattern
try {
    processMessage(record);
} catch (Exception e) {
    log.error("Failed to process offset {}", record.offset(), e);
    deadLetterProducer.send(new ProducerRecord<>("my-topic.DLT",
        record.key(), record.value()));
}
```

Kafka in Code (Java/Spring)

23. How do you manually commit offsets in Spring Kafka?

```
@KafkaListener(topics = "my-topic")
public void listen(ConsumerRecord<String, String> record,
    Acknowledgment ack) {

    try {
        processMessage(record);
        ack.acknowledge(); // commit only after success
    } catch (Exception e) {
        log.error("Processing failed", e);
        // Don't ack - message will be redelivered
    }
}
```

```
}  
}
```

Set `ack-mode: manual` in application config.

24. How do you send a message with a key in Spring Kafka?

```
// Key ensures same-key messages go to same partition  
kafkaTemplate.send("my-topic", userId, payload);
```

Using keys guarantees ordering for the same user/entity across partitions.

25. How do you configure a retry topic in Spring Kafka?

```
@RetryableTopic(  
    attempts = "3",  
    backoff = @Backoff(delay = 1000, multiplier = 2),  
    dltTopicSuffix = ".DLT"  
)  
@KafkaListener(topics = "my-topic")  
public void listen(String message) {  
    processMessage(message); // Auto-retried to my-topic-retry-0, retry-1, then DLT  
}
```

Kafka Connect & Streams (Practical)

26. How do you check status of a Kafka Connect connector?

```
# List connectors  
curl http://localhost:8083/connectors  
  
# Get connector status  
curl http://localhost:8083/connectors/my-connector/status  
  
# Restart a failed connector  
curl -X POST http://localhost:8083/connectors/my-connector/restart
```

27. How do you pause and resume a connector?

```
# Pause  
curl -X PUT http://localhost:8083/connectors/my-connector/pause  
  
# Resume  
curl -X PUT http://localhost:8083/connectors/my-connector/resume
```

28. What do you check when a Kafka Connect connector fails?

- Check connector status via REST API

- Check Connect worker logs for stack traces
- Verify source/sink system availability (DB down? S3 permissions?)
- Check schema compatibility if using Schema Registry
- Verify connector config (wrong hostname, credentials, topic name)

Performance & Capacity Planning

29. How do you decide the number of partitions for a topic? Rule of thumb:

- **Target throughput / single partition throughput** = partition count
- A single partition handles ~10–100 MB/s depending on hardware
- Consider max consumers you'll ever need (partitions $\geq$ max consumers)
- Start with more partitions — you can't reduce later
- Common starting point: `max(target_MB/s ÷ 10, num_consumers)`

30. How do you monitor Kafka health day-to-day? Key metrics to watch:

Metric	Tool	Alert Threshold
Consumer Lag	Kafka UI, Burrow, Prometheus	> 10k messages
Under-replicated Partitions	JMX, Kafka Exporter	> 0
Active Controller Count	JMX	Must be exactly 1
Request latency (produce/fetch)	Prometheus	> 500ms
Disk usage per broker	Grafana	> 80%
ISR Shrink Rate	JMX	Any sudden spike

Popular monitoring stacks: **Prometheus + Grafana + Kafka Exporter** or **Confluent Control Center**.

Real-World Scenarios

31. A downstream service is slow — how do you prevent it from backing up Kafka?

- Use a separate consumer group for the slow service
- Add a buffer queue between Kafka consumer and the slow service
- Scale out consumers and process asynchronously
- Implement backpressure — pause polling when downstream is overwhelmed:

```
consumer.pause(consumer.assignment());
// resume when downstream recovers
consumer.resume(consumer.assignment());
```

32. How do you replay events from a specific time in production?

```
kafka-consumer-groups.sh \
  --bootstrap-server localhost:9092 \
  --group my-group \
  --topic my-topic \
  --reset-offsets \
  --to-datetime 2024-03-01T00:00:00.000 \
  --execute
```

Stop consumers first, reset offset to desired time, restart consumers.

33. How do you safely deploy a new version of a consumer without losing messages?

- Use **rolling deployment** — bring up new instances before taking down old ones
- Since partitions are reassigned within the group, no messages are missed
- Use `group.instance.id` (Static Membership) to reduce rebalance disruption during deploys
- Monitor lag during deployment to confirm no backlog builds up

Want me to cover Kafka security configurations, Schema Registry day-to-day, or Kafka Streams operations in more depth?

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Kafka Scenario-Based Questions & Answers

Data Loss & Durability Scenarios

1. Your team reports that some messages are missing in production. How do you investigate?

Step-by-step approach:

1. Check producer acks config → is it `acks=0` or `acks=1`?
2. Check if any brokers went down recently
3. Check under-replicated partitions
4. Check producer retry config and error logs
5. Compare producer send count vs consumer receive count
6. Check topic retention – messages may have expired

Root causes & fixes:

Cause	Fix
<code>acks=0</code> or <code>acks=1</code>	Set <code>acks=all</code>
No retries on producer	Set <code>retries=Integer.MAX_VALUE</code>
Broker crashed before replication	Ensure <code>min.insync.replicas=2</code>
Messages expired	Increase <code>retention.ms</code>
Consumer committed before processing	Use manual commit after processing

Key broker config to prevent data loss:

```
# Broker level
min.insync.replicas=2
unclean.leader.election.enable=false

# Producer level
acks=all
enable.idempotence=true
retries=Integer.MAX_VALUE
```

2. A broker goes down in a 3-broker cluster. What happens and what do you do?

What Kafka does automatically:

- Controller detects broker failure via ZooKeeper/KRaft heartbeat timeout
- Controller elects new leaders for all partitions that had the failed broker as leader
- Producers/consumers reconnect to new leaders automatically

- ISR shrinks — replication factor temporarily reduced

What you should do:

```
# Step 1: Check under-replicated partitions
kafka-topics.sh --bootstrap-server localhost:9092 \
  --describe --under-replicated-partitions

# Step 2: Check cluster health
kafka-broker-api-versions.sh --bootstrap-server localhost:9092

# Step 3: Check active controller
kafka-metadata-quorum.sh --bootstrap-server localhost:9092 \
  --describe (KRaft mode)

# Step 4: Restart the failed broker
# Step 5: Verify ISR recovers back to full replication
```

Risk window: Between broker failure and recovery, you have reduced replication. If another broker fails during this window, you risk data loss. Prioritize fast recovery.

3. You set `min.insync.replicas=2` but producers are getting `NotEnoughReplicasException`. Why?

This happens when the number of **in-sync replicas drops below 2** — meaning only 1 broker is alive or caught up.

Scenario: 3-broker cluster, replication-factor=3, min.insync.replicas=2. Two brokers go down → only 1 ISR remains → producer with `acks=all` gets rejected.

Fix options:

```
# Option 1: Temporarily lower min.insync.replicas (emergency only)
kafka-configs.sh --alter --bootstrap-server localhost:9092 \
  --entity-type topics --entity-name my-topic \
  --add-config min.insync.replicas=1

# Option 2: Bring failed brokers back online (preferred)

# Option 3: Allow unclean leader election (data loss risk)
unclean.leader.election.enable=true # NOT recommended for critical topics
```

Consumer Issues

4. Your consumer group suddenly stops processing messages. How do you debug?

```
# Step 1: Check if consumers are alive
kafka-consumer-groups.sh --bootstrap-server localhost:9092 \
  --describe --group my-group

# Look at CONSUMER-ID column — empty means no active consumers
```



```

# Step 2: Check lag trend
# Is lag growing? Consumers alive but slow?
# Is lag static? Consumers dead?

# Step 3: Check application logs for exceptions

# Step 4: Check if topic exists and has messages
kafka-console-consumer.sh --bootstrap-server localhost:9092 \
  --topic my-topic --from-beginning --max-messages 3

# Step 5: Check for rebalance loop in logs
# Look for repeated "Rebalance in progress" messages

```

Common causes:

Symptom	Cause	Fix
No active consumers	App crashed	Restart consumers
Consumers active, no progress	Processing deadlock	Check thread dumps
Constant rebalancing	Poll timeout exceeded	Increase <code>max.poll.interval.ms</code>
Deserialization errors	Schema mismatch	Fix schema, route to DLT
Consumer kicked out of group	Session timeout	Tune heartbeat configs

5. Your consumer is processing very slowly and lag keeps growing. What do you do?

Diagnose first:

```

# Check lag per partition
kafka-consumer-groups.sh --bootstrap-server localhost:9092 \
  --describe --group my-group

# Is lag uniform across partitions or concentrated in one?
# Concentrated → that consumer instance is the bottleneck
# Uniform → all consumers are slow

```

Solutions in order of effort:

1. Scale out – add more consumer instances (up to partition count)
2. Increase partitions – allows more consumers
3. Increase `max.poll.records` – process more per batch
4. Make processing async – don't block the poll loop
5. Optimize business logic – DB queries, external calls
6. Use parallel processing within each consumer

Async processing pattern:

```
@KafkaListener(topics = "my-topic")
public void listen(List<ConsumerRecord<String, String>> records) {
    List<CompletableFuture<Void>> futures = records.stream()
        .map(r -> CompletableFuture.runAsync(() -> process(r), executor))
        .collect(toList());

    CompletableFuture.allOf(futures.toArray(new CompletableFuture[0])).join();
    acknowledgment.acknowledge(); // commit after all done
}
```

6. Two consumer groups are reading the same topic — one is fast, one is slow. Does the slow one affect the fast one?

No — consumer groups are completely independent. Each group maintains its own offsets. The slow group will just accumulate lag without affecting the fast group.

However, watch out for:

- **Log retention** — if slow group falls so far behind that Kafka deletes old segments, the slow consumer loses messages
- Fix: Increase `retention.ms` or monitor lag against retention window

```
# Check how far behind the slow group is
kafka-consumer-groups.sh --bootstrap-server localhost:9092 \
    --describe --group slow-group

# Compare lag vs retention window
# If lag offset < earliest available offset → messages already deleted
```

7. A consumer reads the same message multiple times after a restart. Why?

Cause: `enable.auto.commit=true` with a commit interval. Messages processed between the last auto-commit and the crash are reprocessed on restart.

Fix — Manual commit after processing:

```
// application.yml
spring:
  kafka:
    consumer:
      enable-auto-commit: false
    listener:
      ack-mode: manual

// Java
@KafkaListener(topics = "my-topic")
public void listen(ConsumerRecord<String, String> record,
                  Acknowledgment ack) {
    processMessage(record);
```

```
ack.acknowledge(); // Only commit AFTER successful processing
}
```

Also make processing idempotent as a safety net:

```
// Check if already processed using a DB/Redis dedup store
if (deduplicationStore.alreadyProcessed(record.offset(), record.partition())) {
    ack.acknowledge();
    return;
}
processMessage(record);
deduplicationStore.markProcessed(record.offset(), record.partition());
ack.acknowledge();
```

Ordering & Partitioning Scenarios

8. You need to guarantee message ordering for events related to the same user. How do you implement this?

Solution: Use `userId` as the partition key. Kafka guarantees order within a partition.

```
// Producer — always send with userId as key
kafkaTemplate.send("user-events", userId, eventPayload);

// All events for the same userId → same partition → ordered
```

Pitfalls to avoid:

- Don't use `null` key — routes round-robin, breaks ordering
- Don't increase partitions without considering key redistribution
- Don't have multiple producers for same user without coordination
- Don't use async processing within a consumer — it breaks partition-level ordering

For strict ordering with multiple consumers:

```
Partition 0 → Consumer A (handles users 0-33%)
Partition 1 → Consumer B (handles users 33-66%)
Partition 2 → Consumer C (handles users 66-100%)
```

Each consumer processes its partition sequentially.

9. After increasing partitions from 3 to 6, some users' events are out of order. Why?

Cause: Kafka uses `hash(key) % numPartitions` to assign partitions. When partition count changes, the same key maps to a **different partition**. Events already in old partitions and new events in new partitions get consumed out of order.

How to handle:

Option 1: Drain old partitions completely before producing new messages
Option 2: Use a consumer-side sequencing mechanism (sequence numbers in payload)
Option 3: Plan partition count upfront to avoid increasing later
Option 4: Use a Kafka Streams reordering buffer with event timestamps

Best practice: Over-provision partitions upfront (e.g., 12 or 24) so you rarely need to increase.

10. You have a topic with 10 partitions but only 3 consumers. How are partitions assigned?

With default **Range** or **RoundRobin** assignor:

```
Consumer A → Partitions 0, 1, 2, 3
Consumer B → Partitions 4, 5, 6
Consumer C → Partitions 7, 8, 9
```

Consumer A handles more partitions and may become a bottleneck. Better to use **Sticky Assignor** which balances evenly and minimizes reassignment on rebalance.

```
partition.assignment.strategy=org.apache.kafka.clients.consumer.StickyAssignor
```

Ideal setup: Number of consumers = Number of partitions for maximum parallelism.

Production Incidents

11. Kafka is receiving 10x normal traffic spike. What do you do?

Immediate actions:

1. Check broker CPU, disk I/O, network throughput
2. Check producer send latency and error rates
3. Check consumer lag – is it growing?
4. Check if any broker is the hotspot (uneven partition distribution)

Short-term mitigations:

```
# Producer side – enable compression
compression.type=snappy # reduces network/disk load by 60-80%
linger.ms=20             # larger batches
batch.size=65536
```

```
# Consumer side – process faster
max.poll.records=1000
```

Longer term:

- Add more brokers and rebalance partitions
- Increase partitions on hot topics

- Enable tiered storage to offload old data
- Use quotas to throttle noisy producers

12. Your Kafka topic is filling up disk space rapidly. How do you handle it?

```
# Step 1: Check topic size per partition
kafka-log-dirs.sh --bootstrap-server localhost:9092 \
  --topic-list my-topic --describe

# Step 2: Check current retention config
kafka-configs.sh --bootstrap-server localhost:9092 \
  --entity-type topics --entity-name my-topic --describe

# Step 3: Reduce retention immediately (emergency)
kafka-configs.sh --alter --bootstrap-server localhost:9092 \
  --entity-type topics --entity-name my-topic \
  --add-config retention.ms=3600000 # 1 hour

# Step 4: Reduce retention by size
kafka-configs.sh --alter --bootstrap-server localhost:9092 \
  --entity-type topics --entity-name my-topic \
  --add-config retention.bytes=1073741824 # 1GB per partition
```

Long-term fixes:

- Enable **log compaction** for changelog-type topics
- Archive old data to S3 using Kafka Connect S3 Sink
- Enable **Tiered Storage** (Kafka 3.6+) to offload to object storage
- Add more broker disk capacity

13. A critical consumer is stuck because of one bad (poison pill) message. How do you skip it?

```
# Step 1: Identify the bad offset
kafka-consumer-groups.sh --bootstrap-server localhost:9092 \
  --describe --group my-group
# Note the stuck partition and current offset

# Step 2: Inspect the message
kafka-console-consumer.sh \
  --bootstrap-server localhost:9092 \
  --topic my-topic \
  --partition 2 \
  --offset 1045 \
  --max-messages 1

# Step 3: Skip the bad message by moving offset forward
```

```
kafka-consumer-groups.sh --bootstrap-server localhost:9092 \  
  --group my-group \  
  --topic my-topic:2 # partition 2  
  --reset-offsets \  
  --shift-by 1 \      # skip 1 message  
  --execute
```

Code-level fix — Dead Letter Topic:

```
@KafkaListener(topics = "my-topic")  
public void listen(ConsumerRecord<String, String> record, Acknowledgment ack) {  
    try {  
        processMessage(record);  
    } catch (DeserializationException e) {  
        // Route to DLT and continue  
        dltTemplate.send("my-topic.DLT", record.key(), record.value());  
        log.error("Poison pill at offset {}, sent to DLT", record.offset());  
    } finally {  
        ack.acknowledge(); // Always commit to move past bad message  
    }  
}
```

14. Two microservices are producing to the same topic with different message formats. Consumers are breaking. How do you fix this?

Root cause: No schema enforcement — any producer can publish any format.

Fix — Enforce schemas with Schema Registry:

```
// Producer config  
props.put("schema.registry.url", "http://schema-registry:8081");  
props.put("value.serializer", KafkaAvroSerializer.class);  
  
// Schema evolution rules (set in Schema Registry)  
// BACKWARD — new schema can read old data (add optional fields)  
// FORWARD  — old schema can read new data  
// FULL     — both directions (safest)
```

Short-term fix without Schema Registry:

```
// Consumer — handle multiple versions  
public void processMessage(String payload) {  
    JsonNode node = objectMapper.readTree(payload);  
    String version = node.has("version") ? node.get("version").asText() : "v1";  
  
    switch (version) {  
        case "v1": handleV1(node); break;  
        case "v2": handleV2(node); break;  
        default: dltProducer.send("my-topic.DLT", payload);  
    }  
}
```

```
}  
}
```

Rebalancing & Stability

15. Your consumers are rebalancing every few minutes disrupting processing. How do you fix it?

Diagnose:

```
# Look for rebalance logs  
grep "Rebalance" application.log | tail -50  
  
# Check if consumers are being evicted  
grep "Member .* in group .* has failed" kafka-server.log
```

Fix 1 — Tune poll and heartbeat timeouts:

```
# Consumer config  
max.poll.interval.ms=600000    # Give 10 min to process a batch  
session.timeout.ms=60000      # More tolerance for slow GC  
heartbeat.interval.ms=10000    # Heartbeat every 10s (1/3 of session timeout)  
max.poll.records=100          # Reduce records per poll if processing is slow
```

Fix 2 — Static Group Membership (best for stable deployments):

```
# Each consumer gets a stable identity – no rebalance on restart  
group.instance.id=consumer-instance-1
```

```
Without static membership: Restart → leave group → rebalance → rejoin → rebalance  
With static membership:   Restart → rejoin with same ID → no rebalance
```

16. You deployed a new consumer version and now messages are being processed twice. Why?

Cause: During rolling deployment, old and new consumers briefly coexist in the same group → rebalance → some partitions are temporarily processed by both.

Fix:

1. Use **Static Group Membership** to minimize rebalance duration
2. Ensure idempotent processing (same message twice = same **result**)
3. Use Blue-Green deployment:
 - **New group** (my-group-v2) starts consuming **from** latest
 - **Old group** (my-group-v1) drains remaining messages
 - Switch traffic, decommission **old group**

Kafka Connect Scenarios

17. Your Kafka Connect JDBC source connector stops pulling new rows. What do you check?

```
# Step 1: Check connector status
curl http://localhost:8083/connectors/jdbc-source/status

# Step 2: Check task status (tasks do the actual work)
curl http://localhost:8083/connectors/jdbc-source/tasks/0/status

# Step 3: Restart failed task
curl -X POST http://localhost:8083/connectors/jdbc-source/tasks/0/restart
```

Common causes:

- Incrementing **column** (id/timestamp) **not** updating → **no new rows** detected
- DB connection lost → task fails silently
- **Table** schema changed → serialization error
- poll.interval.ms too high → infrequent polling
- **Offset** stored **for** a deleted/truncated **table** → reset needed

Reset connector offsets:

```
# Delete stored offsets from __consumer_offsets or offset topic
# Then restart connector to re-read from beginning
curl -X DELETE http://localhost:8083/connectors/jdbc-source
# Recreate with reset config
```

18. Messages are appearing out of order in your S3 sink connector output files. How do you fix it?

Cause: Multiple tasks write to S3 in parallel — different partitions may flush at different rates.

Fix — Enforce ordering in S3 output:

```
{
  "name": "s3-sink",
  "config": {
    "tasks.max": "1",
    "flush.size": "1000",
    "rotate.interval.ms": "60000",
    "locale": "en_US",
    "timezone": "UTC",
    "timestamp.extractor": "RecordField",
    "timestamp.field": "event_timestamp",
    "s3.part.size": "67108864",
    "storage.class": "io.confluent.connect.s3.storage.S3Storage",
    "format.class": "io.confluent.connect.s3.format.parquet.ParquetFormat",
    "partitioner.class": "io.confluent.connect.storage.partitioners.TimeBasedPartitioner",
```



```
    "path.format": "'year'='YYYY/'month'='MM/'day'='dd/'hour'='HH'"
  }
}
```

For strict ordering, sort downstream in your data warehouse/lake on the event timestamp field.

Exactly-Once & Transactions

19. You need to read from one Kafka topic, transform data, and write to another — with no duplicates even on failure. How?

Solution — Kafka Streams with exactly-once:

```
Properties props = new Properties();
props.put(StreamsConfig.PROCESSING_GUARANTEE_CONFIG,
    StreamsConfig.EXACTLY_ONCE_V2); // Kafka 2.5+

StreamsBuilder builder = new StreamsBuilder();
builder.stream("input-topic")
    .mapValues(value -> transform(value))
    .to("output-topic");

KafkaStreams streams = new KafkaStreams(builder.build(), props);
streams.start();
```

Solution — Manual transactions (producer-consumer):

```
producer.initTransactions();
try {
    consumer.poll(Duration.ofMillis(100)).forEach(record -> {
        producer.beginTransaction();
        producer.send(new ProducerRecord<>("output-topic",
            record.key(), transform(record.value())));
        // Commit consumer offset as part of transaction
        producer.sendOffsetsToTransaction(
            Collections.singletonMap(
                new TopicPartition(record.topic(), record.partition()),
                new OffsetAndMetadata(record.offset() + 1)),
            consumer.groupMetadata());
        producer.commitTransaction();
    });
} catch (Exception e) {
    producer.abortTransaction();
}
```

20. Your transactional producer keeps failing with `ProducerFencedException`. Why?

Cause: Two producer instances are using the **same** `transactional.id`. Kafka fences out the older instance when a new one initializes with the same ID — this is a safety mechanism.

Fix:

```
// Each producer instance must have a UNIQUE transactional.id
props.put("transactional.id", "my-producer-" + instanceId);
// instanceId should be unique per pod/process

// In Kubernetes – use pod name as part of transactional.id
String txId = "my-service-" + System.getenv("HOSTNAME");
props.put("transactional.id", txId);
```

Multi-Datacenter & Disaster Recovery

21. How do you set up Kafka replication across two data centers?

Tool: MirrorMaker 2 (MM2)

```
# mm2.properties
clusters = primary, secondary
primary.bootstrap.servers = primary-kafka:9092
secondary.bootstrap.servers = secondary-kafka:9092

# Replicate primary → secondary
primary->secondary.enabled = true
primary->secondary.topics = .* # all topics

# Topic naming on secondary: primary.my-topic
replication.factor = 2
```

```
# Start MirrorMaker 2
connect-mirror-maker.sh mm2.properties
```

Active-Active vs Active-Passive:

```
Active-Passive: Primary handles all traffic, secondary is DR only
Active-Active: Both datacenters produce/consume, MM2 syncs both ways
                (requires careful handling of replication cycles)
```

22. Your MirrorMaker2 replication is lagging behind. How do you fix it?

```
# Check replication lag
kafka-consumer-groups.sh \
  --bootstrap-server secondary-kafka:9092 \
  --describe --group mm2-group
```

```
# Check MM2 metrics via JMX or Prometheus
# Key metric: replication.latency.ms
```

Fixes:

```
# Increase tasks for higher throughput
tasks.max = 8
```

```
# Tune replication flow
replication.factor = 2
refresh.topics.interval.seconds = 30
sync.topic.configs.enabled = true
```

```
# Increase MM2 heap if OOM
export KAFKA_HEAP_OPTS="-Xms2g -Xmx4g"
```

Want me to go deeper into any specific area — like Kafka Streams stateful processing scenarios, Schema Registry conflicts, or Kafka security incident scenarios?

5/2/2026, 11:16:54 PM

Kafka Topics We Haven't Covered Yet

Quick Overview of Missing Areas

Here's a comprehensive map of what we haven't covered:

1. Kafka Internals Deep Dive

How does Kafka's log segment work internally?

Each partition is stored as a set of **segment files** on disk:

```
/kafka-logs/my-topic-0/
├─ 00000000000000000000.log      # actual message data
├─ 00000000000000000000.index    # offset → file position index
├─ 00000000000000000000.timeindex # timestamp → offset index
├─ 000000000000000000001234567.log # next segment
└─ leader-epoch-checkpoint
```

How a message is written:

```
Producer → Leader Broker
→ Appended to active .log segment (sequential write)
→ Index updated every index.interval.bytes
→ Flushed to OS page cache (not necessarily disk)
→ Followers fetch and replicate
→ Leader updates ISR
→ Ack sent to producer
```

Segment rolling — when does a new segment start?

```
# New segment created when any of these hit:
log.segment.bytes=1073741824 # segment reaches 1GB
log.roll.ms=604800000        # segment is 7 days old
log.roll.jitter.ms=0         # jitter to avoid all segments rolling at once
log.index.size.max.bytes=10485760 # index file reaches 10MB
```

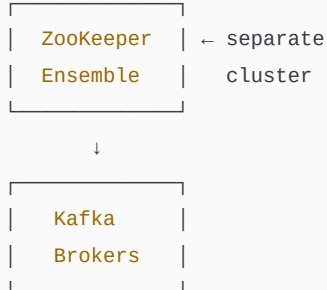
Why sequential writes make Kafka fast:

```
Random disk write → ~100 IOPS (HDD)
Sequential write → ~3000 MB/s (HDD), much faster on SSD
Kafka always appends → always sequential → very fast
```

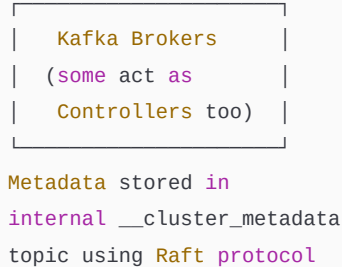
2. KRaft Mode (Kafka Without ZooKeeper)

What changed with KRaft (Kafka 3.3+ production ready):

Before KRaft:



After KRaft:



KRaft roles:

```
# Broker only
process.roles=broker
```

```
# Controller only (manages metadata)
process.roles=controller
```

```
# Combined (good for small clusters)
process.roles=broker,controller
```

Check KRaft quorum status:

```
kafka-metadata-quorum.sh \
  --bootstrap-server localhost:9092 \
  --describe
```

Benefits of KRaft:

- Faster controller failover (milliseconds vs seconds)
- Supports millions of partitions (ZK had ~200K limit)
- Simpler operations — one less system to manage
- Better metadata consistency

3. Kafka Security In Depth

SSL/TLS Setup:

```
# Generate broker keystore
keytool -keystore kafka.server.keystore.jks \
  -alias localhost -keyalg RSA -validity 365 \
  -genkey -storepass password

# Generate truststore
```

```
keytool -keystore kafka.server.truststore.jks \  
-alias CARoot -import -file ca-cert \  
-storepass password
```

Broker config

```
listeners=SSL://localhost:9093  
ssl.keystore.location=/path/to/kafka.server.keystore.jks  
ssl.keystore.password=password  
ssl.key.password=password  
ssl.truststore.location=/path/to/kafka.server.truststore.jks  
ssl.truststore.password=password  
ssl.client.auth=required # mutual TLS
```

SASL Authentication types:

```
SASL/PLAIN    → username/password (simple, use with SSL)  
SASL/SCRAM    → salted passwords stored in ZK/KRaft  
SASL/GSSAPI   → Kerberos (enterprise environments)  
SASL/OAUTHBEARER → OAuth2 tokens (cloud-native)
```

ACL Management:

Grant producer access

```
kafka-acls.sh --bootstrap-server localhost:9092 \  
--add --allow-principal User:service-a \  
--operation Write --topic my-topic
```

Grant consumer access

```
kafka-acls.sh --bootstrap-server localhost:9092 \  
--add --allow-principal User:service-b \  
--operation Read --topic my-topic \  
--group my-consumer-group
```

List ACLs

```
kafka-acls.sh --bootstrap-server localhost:9092 \  
--list --topic my-topic
```

Producer/Consumer Quotas:

Throttle a producer to 10MB/s

```
kafka-configs.sh --bootstrap-server localhost:9092 \  
--alter --entity-type clients \  
--entity-name my-producer \  
--add-config producer_byte_rate=10485760
```

Throttle consumer

```
kafka-configs.sh --bootstrap-server localhost:9092 \  
--alter --entity-type clients \  
--entity-name my-consumer \  
--add-config consumer_byte_rate=10485760
```

4. Schema Registry Deep Dive

Schema compatibility modes explained:

BACKWARD → New schema can read data written with old schema
(consumers upgrade first)
Safe change: add optional field with default

FORWARD → Old schema can read data written with new schema
(producers upgrade first)
Safe change: remove optional field

FULL → Both BACKWARD and FORWARD
Safest – only add/remove optional fields with defaults

NONE → No compatibility check (dangerous in production)

Avro schema evolution example:

```
// v1 schema
{
  "type": "record",
  "name": "User",
  "fields": [
    {"name": "id", "type": "string"},
    {"name": "name", "type": "string"}
  ]
}

// v2 schema – BACKWARD compatible (added optional field)
{
  "type": "record",
  "name": "User",
  "fields": [
    {"name": "id", "type": "string"},
    {"name": "name", "type": "string"},
    {"name": "email", "type": ["null", "string"], "default": null}
  ]
}
```

Schema Registry API:

```
# Register schema
curl -X POST http://schema-registry:8081/subjects/my-topic-value/versions \
  -H "Content-Type: application/json" \
  -d '{"schema": "{\"type\": \"string\"}'}'

# Get all versions
curl http://schema-registry:8081/subjects/my-topic-value/versions
```

```
# Check compatibility before registering
curl -X POST http://schema-registry:8081/compatibility/subjects/my-topic-value/versions/latest \
  -H "Content-Type: application/json" \
  -d @new-schema.json
```

5. Kafka Streams Deep Dive

Stateful operations — most asked in interviews:

```
StreamsBuilder builder = new StreamsBuilder();

// Count events per user in 5-minute windows
KStream<String, String> stream = builder.stream("events");

stream
  .groupByKey()
  .windowedBy(TimeWindows.ofSizeWithNoGrace(Duration.ofMinutes(5)))
  .count()
  .toStream()
  .to("event-counts");
```

Joins in Kafka Streams:

```
// Stream-Stream Join (both must be within join window)
KStream<String, Order> orders = builder.stream("orders");
KStream<String, Payment> payments = builder.stream("payments");

orders.join(
  payments,
  (order, payment) -> new OrderPayment(order, payment),
  JoinWindows.ofTimeDifferenceWithNoGrace(Duration.ofMinutes(5)),
  StreamJoined.with(Serdes.String(), orderSerde, paymentSerde)
).to("order-payments");

// Stream-Table Join (table is always latest state)
KTable<String, UserProfile> profiles = builder.table("user-profiles");
KStream<String, Event> events = builder.stream("events");

events.join(
  profiles,
  (event, profile) -> enrich(event, profile)
).to("enriched-events");

// Table-Table Join
KTable<String, Order> orderTable = builder.table("orders");
KTable<String, Customer> customerTable = builder.table("customers");

orderTable.join(
```



```
customerTable,
    (order, customer) -> new EnrichedOrder(order, customer)
).toStream().to("enriched-orders");
```

State Stores — for custom stateful processing:

```
// Create a persistent state store
StoreBuilder<KeyValueStore<String, Long>> storeBuilder =
    Stores.keyValueStoreBuilder(
        Stores.persistentKeyValueStore("my-state-store"),
        Serdes.String(),
        Serdes.Long()
    );
builder.addStateStore(storeBuilder);

// Use in a processor
stream.process(() -> new Processor<String, String, String, Long>() {
    private KeyValueStore<String, Long> store;

    @Override
    public void init(ProcessorContext<String, Long> context) {
        store = context.getStateStore("my-state-store");
    }

    @Override
    public void process(Record<String, String> record) {
        Long count = store.get(record.key());
        store.put(record.key(), count == null ? 1 : count + 1);
    }
}, "my-state-store");
```

Interactive Queries — query state stores from outside:

```
// Query local state store
ReadOnlyKeyValueStore<String, Long> store =
    streams.store(StoreQueryParameters.fromNameAndType(
        "my-state-store", QueryableStoreTypes.keyValueStore()));

Long count = store.get("user-123");

// For distributed queries across instances
// Use streams.metadataForKey() to find which instance owns a key
```

6. Tiered Storage (Kafka 3.6+)

Offloads old log segments to cheap object storage (S3, GCS, Azure Blob) while keeping recent data on broker disk.

```
# Enable tiered storage
remote.log.storage.system.enable=true
```

```
remote.log.storage.manager.class.name=org.apache.kafka.server.log.remote.storage.RemoteLogStorageManager

# Keep only last 1 day on local disk
log.retention.ms=86400000
# But keep 30 days in remote storage
remote.log.metadata.manager.impl.prefix=rlmm.config.
remote.log.metadata.topic.replication.factor=3
```

Benefits:

Without tiered storage:

Broker disk → must **hold ALL** data **within** retention **period**
Large retention = expensive broker disks

With tiered storage:

Broker disk → holds **only** recent data (e.g., **last 1 day**)
S3/GCS → holds older data cheaply
Consumers → transparently read **from** either location

7. Kafka Performance Tuning Checklist

Broker-level tuning:

```
# Network threads (handles network requests)
num.network.threads=8

# IO threads (handles disk reads/writes)
num.io.threads=16

# Socket buffers
socket.send.buffer.bytes=1048576
socket.receive.buffer.bytes=1048576
socket.request.max.bytes=104857600

# Log flushing (rely on OS – don't flush too often)
log.flush.interval.messages=Long.MAX_VALUE
log.flush.interval.ms=Long.MAX_VALUE

# Replication throttle for non-critical topics
replica.fetch.max.bytes=10485760
```

OS-level tuning:

```
# Increase file descriptors
ulimit -n 100000

# Tune virtual memory
sysctl -w vm.swappiness=1
sysctl -w vm.dirty_background_ratio=5
sysctl -w vm.dirty_ratio=60
```

```
# Network buffers
sysctl -w net.core.rmem_max=134217728
sysctl -w net.core.wmem_max=134217728

# Use XFS filesystem for Kafka logs (better performance)
mkfs.xfs /dev/sdb
mount -o noatime /dev/sdb /kafka-logs
```

8. Kafka Testing

Unit testing with EmbeddedKafka (Spring):

```
@SpringBootTest
@EmbeddedKafka(
    partitions = 3,
    topics = {"input-topic", "output-topic"},
    brokerProperties = {"auto.create.topics.enable=false"}
)
class KafkaIntegrationTest {

    @Autowired
    private KafkaTemplate<String, String> kafkaTemplate;

    @Test
    void testMessageProcessing() throws Exception {
        kafkaTemplate.send("input-topic", "key1", "test-message");

        // Use CountdownLatch or Awaitility to wait for processing
        await().atMost(10, SECONDS)
            .untilAsserted(() ->
                assertThat(outputMessages).contains("processed-test-message")
            );
    }
}
```

Testing with Testcontainers (closer to production):

```
@Testcontainers
class KafkaContainerTest {

    @Container
    static KafkaContainer kafka = new KafkaContainer(
        DockerImageName.parse("confluentinc/cp-kafka:7.5.0")
    );

    @Test
    void testProduceAndConsume() {
        String bootstrapServers = kafka.getBootstrapServers();
```

```

        // Create producer/consumer with real Kafka instance
        // Full integration test with real broker behavior
    }
}

```

Kafka Topology Test Driver (Kafka Streams unit test):

```

@Test
void testStreamTransformation() {
    TopologyTestDriver driver = new TopologyTestDriver(topology, props);

    TestInputTopic<String, String> inputTopic = driver.createInputTopic(
        "input", new StringSerializer(), new StringSerializer());

    TestOutputTopic<String, String> outputTopic = driver.createOutputTopic(
        "output", new StringDeserializer(), new StringDeserializer());

    inputTopic.pipeInput("key1", "hello");

    assertThat(outputTopic.readValue()).isEqualTo("HELLO");
    driver.close();
}

```

9. Kafka Observability Stack

Full monitoring setup:

```

# docker-compose snippet
services:
  kafka-exporter:
    image: danielqsj/kafka-exporter
    command: --kafka.server=kafka:9092

  prometheus:
    image: prom/prometheus
    volumes:
      - ./prometheus.yml:/etc/prometheus/prometheus.yml

  grafana:
    image: grafana/grafana
    # Import dashboard ID 7589 for Kafka overview

```

Key JMX metrics to expose:

```

# Broker health
kafka.server:type=BrokerTopicMetrics,name=MessagesInPerSec
kafka.server:type=BrokerTopicMetrics,name=BytesInPerSec
kafka.server:type=ReplicaManager,name=UnderReplicatedPartitions

```

```
kafka.controller:type=KafkaController,name=ActiveControllerCount

# Producer metrics
kafka.producer:type=producer-metrics,name=record-error-rate
kafka.producer:type=producer-metrics,name=request-latency-avg

# Consumer metrics
kafka.consumer:type=consumer-fetch-manager-metrics,name=records-lag-max
```

Alerting rules (Prometheus):

```
groups:
- name: kafka_alerts
  rules:
    - alert: KafkaConsumerLagHigh
      expr: kafka_consumergroup_lag > 10000
      for: 5m
      annotations:
        summary: "Consumer lag too high"

    - alert: KafkaUnderReplicatedPartitions
      expr: kafka_server_replicamanager_underreplicatedpartitions > 0
      for: 1m
      annotations:
        summary: "Under-replicated partitions detected"

    - alert: KafkaNoActiveController
      expr: kafka_controller_kafkacontroller_activecontrollercount != 1
      for: 1m
      annotations:
        summary: "Kafka has no active controller"
```

10. Kafka Design Patterns

Event Sourcing with Kafka:

Every state change = an event published to Kafka
Current state = replay of all events from beginning

```
User created → { type: USER_CREATED, userId: 1, name: "John" }
Email updated → { type: EMAIL_UPDATED, userId: 1, email: "j@j.com" }
Account closed → { type: ACCOUNT_CLOSED, userId: 1 }
```

Rebuild state anytime by replaying from offset 0

CQRS with Kafka:

Command side → Validates + produces events to Kafka
Query side → Consumes events + builds read-optimized views

Orders Service → produces to "orders" topic
Inventory View → consumes "orders" → builds inventory read model
Analytics View → consumes "orders" → builds reporting model

Saga Pattern with Kafka:

Order Service → publishes OrderCreated
Payment Service → consumes OrderCreated → publishes PaymentProcessed
Inventory Service → consumes PaymentProcessed → publishes InventoryReserved
Shipping Service → consumes InventoryReserved → publishes OrderShipped

On failure:
Payment fails → publishes PaymentFailed
Order Service → consumes PaymentFailed → cancels order (compensating tx)

Outbox Pattern (prevents dual-write problem):

```
// WRONG - dual write, not atomic
orderRepository.save(order);           // DB write
kafkaTemplate.send("orders", order);   // Kafka write
// If Kafka fails after DB save → inconsistency

// RIGHT - Outbox pattern
@Transactional
public void createOrder(Order order) {
    orderRepository.save(order);
    // Write to outbox table in SAME transaction
    outboxRepository.save(new OutboxEvent("orders", order));
}
// Separate process (Debezium CDC) reads outbox table
// and publishes to Kafka - guaranteed consistency
```

Summary of All Topics Covered Across Sessions

Area	Covered
Core concepts & architecture	✓
CLI operations	✓
Producer/Consumer configs	✓
Consumer lag & monitoring	✓
Scenario-based debugging	✓
Kafka Internals & log segments	✓

Area	Covered
KRaft mode	✓
Security (SSL, SASL, ACLs)	✓
Schema Registry	✓
Kafka Streams (stateful, joins)	✓
Tiered Storage	✓
Performance tuning	✓
Testing strategies	✓
Observability & alerting	✓
Design patterns	✓
Multi-datacenter / MirrorMaker2	✓
Want me to create a PDF study guide combining all Kafka Q&A from all three sessions?	

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Kafka Certificates, Serialization & Configuration

PART 1: Connecting to Kafka with Certificates (SSL/TLS)

How SSL Works in Kafka

```
Client (Producer/Consumer)
  ↓ presents client certificate
Kafka Broker
  ↓ validates against truststore
Mutual TLS (mTLS) established
  ↓ encrypted communication
Messages flow securely
```

Certificate Files You'll Typically Receive

kafka-ca.crt	→ Certificate Authority (CA) cert
client.crt	→ Your client certificate
client.key	→ Your private key
client.p12 / .pfx	→ Combined cert+key (PKCS12 format)
kafka.truststore.jks	→ Java truststore (contains CA cert)
kafka.keystore.jks	→ Java keystore (contains client cert+key)

Step 1 — Convert Certificates to JKS (if given .crt/.key files)

```
# Step 1: Create PKCS12 from cert + key
openssl pkcs12 -export \
  -in client.crt \
  -inkey client.key \
  -out client.p12 \
  -name kafka-client \
  -passout pass:yourpassword

# Step 2: Import PKCS12 into Java Keystore
keytool -importkeystore \
  -deststorepass yourpassword \
  -destkeypass yourpassword \
  -destkeystore kafka.keystore.jks \
  -srckeystore client.p12 \
```



```
-srcstoretype PKCS12 \  
-srcstorepass yourpassword \  
-alias kafka-client  
  
# Step 3: Create truststore from CA cert  
keytool -import \  
-trustcacerts \  
-alias kafka-ca \  
-file kafka-ca.crt \  
-keystore kafka.truststore.jks \  
-storepass yourpassword \  
-noprompt  
  
# Verify keystore contents  
keytool -list -v -keystore kafka.keystore.jks -storepass yourpassword  
keytool -list -v -keystore kafka.truststore.jks -storepass yourpassword
```

Step 2 — Java Producer/Consumer with SSL

```
Properties props = new Properties();  
  
// Basic connection  
props.put("bootstrap.servers", "kafka-broker:9093");  
  
// SSL settings  
props.put("security.protocol", "SSL");  
props.put("ssl.truststore.location", "/certs/kafka.truststore.jks");  
props.put("ssl.truststore.password", "yourpassword");  
props.put("ssl.keystore.location", "/certs/kafka.keystore.jks");  
props.put("ssl.keystore.password", "yourpassword");  
props.put("ssl.key.password", "yourpassword");  
  
// Optional: disable hostname verification (not recommended for prod)  
// props.put("ssl.endpoint.identification.algorithm", "");  
  
// Serializers (for producer)  
props.put("key.serializer", StringSerializer.class.getName());  
props.put("value.serializer", StringSerializer.class.getName());  
  
KafkaProducer<String, String> producer = new KafkaProducer<>(props);
```

Step 3 — Spring Boot with SSL

application.yml:

```
spring:  
  kafka:
```

```
bootstrap-servers: kafka-broker:9093
properties:
  security.protocol: SSL
  ssl.truststore.location: /certs/kafka.truststore.jks
  ssl.truststore.password: yourpassword
  ssl.keystore.location: /certs/kafka.keystore.jks
  ssl.keystore.password: yourpassword
  ssl.key.password: yourpassword
producer:
  key-serializer: org.apache.kafka.common.serialization.StringSerializer
  value-serializer: org.apache.kafka.common.serialization.StringSerializer
consumer:
  key-deserializer: org.apache.kafka.common.serialization.StringDeserializer
  value-deserializer: org.apache.kafka.common.serialization.StringDeserializer
  group-id: my-group
```

SASL + SSL (Most Common in Enterprise)

```
# application.yml – SASL/SCRAM with SSL
spring:
  kafka:
    bootstrap-servers: kafka-broker:9093
    properties:
      security.protocol: SASL_SSL
      sasl.mechanism: SCRAM-SHA-512
      sasl.jaas.config: >
        org.apache.kafka.common.security.scram.ScramLoginModule required
        username="my-service"
        password="mypassword";
    ssl.truststore.location: /certs/kafka.truststore.jks
    ssl.truststore.password: yourpassword
```

```
# SASL/PLAIN with SSL
spring:
  kafka:
    properties:
      security.protocol: SASL_SSL
      sasl.mechanism: PLAIN
      sasl.jaas.config: >
        org.apache.kafka.common.security.plain.PlainLoginModule required
        username="my-service"
        password="mypassword";
```

```
# Kerberos (GSSAPI) – enterprise environments
spring:
  kafka:
    properties:
      security.protocol: SASL_SSL
      sasl.mechanism: GSSAPI
```

```
sasl.kerberos.service.name: kafka
sasl.jaas.config: >
    com.sun.security.auth.module.Krb5LoginModule required
    useKeyTab=true
    storeKey=true
    keyTab="/etc/security/kafka.keytab"
    principal="kafka-client@REALM.COM";
```

SSL with Python (kafka-python / confluent-kafka)

```
# kafka-python
from kafka import KafkaProducer, KafkaConsumer

producer = KafkaProducer(
    bootstrap_servers='kafka-broker:9093',
    security_protocol='SSL',
    ssl_cafile='/certs/kafka-ca.crt',
    ssl_certfile='/certs/client.crt',
    ssl_keyfile='/certs/client.key',
    ssl_password='yourpassword'
)

consumer = KafkaConsumer(
    'my-topic',
    bootstrap_servers='kafka-broker:9093',
    security_protocol='SSL',
    ssl_cafile='/certs/kafka-ca.crt',
    ssl_certfile='/certs/client.crt',
    ssl_keyfile='/certs/client.key',
    group_id='my-group'
)

# confluent-kafka (recommended for production)
from confluent_kafka import Producer, Consumer

conf = {
    'bootstrap.servers': 'kafka-broker:9093',
    'security.protocol': 'SSL',
    'ssl.ca.location': '/certs/kafka-ca.crt',
    'ssl.certificate.location': '/certs/client.crt',
    'ssl.key.location': '/certs/client.key',
    'ssl.key.password': 'yourpassword'
}

producer = Producer(conf)
consumer = Consumer(**conf, 'group.id': 'my-group',
                    'auto.offset.reset': 'earliest')
```

Debugging SSL Connection Issues

```
# Test SSL connection from terminal
openssl s_client \
  -connect kafka-broker:9093 \
  -CAfile kafka-ca.crt \
  -cert client.crt \
  -key client.key

# Verify certificate details
openssl x509 -in client.crt -text -noout

# Check certificate expiry
openssl x509 -enddate -noout -in client.crt

# Test with kafka CLI
kafka-console-producer.sh \
  --bootstrap-server kafka-broker:9093 \
  --topic test \
  --producer.config ssl-client.properties
```

```
# ssl-client.properties
security.protocol=SSL
ssl.truststore.location=/certs/kafka.truststore.jks
ssl.truststore.password=yourpassword
ssl.keystore.location=/certs/kafka.keystore.jks
ssl.keystore.password=yourpassword
ssl.key.password=yourpassword
```

PART 2: Serialization in Kafka

Why Serialization Matters

```
Producer side:  Java Object → bytes  (Serialization)
Kafka stores:   bytes  only
Consumer side:  bytes  → Java Object (Deserialization)

Wrong serializer/deserializer = SerializationException → consumer crashes
```

Built-in Serializers

```
// All available out of the box
StringSerializer / StringDeserializer
ByteArraySerializer / ByteArrayDeserializer
IntegerSerializer / IntegerDeserializer
LongSerializer / LongDeserializer
DoubleSerializer / DoubleDeserializer
```

```
FloatSerializer / FloatDeserializer
UUIDSerializer / UUIDDeserializer

// Usage
props.put("key.serializer", StringSerializer.class.getName());
props.put("value.serializer", LongSerializer.class.getName());
```

JSON Serialization

```
// Custom JSON Serializer
public class JsonSerializer<T> implements Serializer<T> {
    private final ObjectMapper objectMapper = new ObjectMapper();

    @Override
    public byte[] serialize(String topic, T data) {
        if (data == null) return null;
        try {
            return objectMapper.writeValueAsBytes(data);
        } catch (JsonProcessingException e) {
            throw new SerializationException("Error serializing JSON", e);
        }
    }
}

// Custom JSON Deserializer
public class JsonDeserializer<T> implements Deserializer<T> {
    private final ObjectMapper objectMapper = new ObjectMapper();
    private Class<T> targetType;

    public JsonDeserializer(Class<T> targetType) {
        this.targetType = targetType;
    }

    @Override
    public T deserialize(String topic, byte[] data) {
        if (data == null) return null;
        try {
            return objectMapper.readValue(data, targetType);
        } catch (IOException e) {
            throw new SerializationException("Error deserializing JSON", e);
        }
    }
}
```

Spring Boot JSON config:

```
spring:
  kafka:
    producer:
```

```
value-serializer: org.springframework.kafka.support.serializer.JsonSerializer
consumer:
  value-deserializer: org.springframework.kafka.support.serializer.JsonDeserializer
  properties:
    spring.json.trusted.packages: "com.myapp.model"
    spring.json.value.default.type: "com.myapp.model.OrderEvent"
```

```
// Spring Kafka listener with JSON
@KafkaListener(topics = "orders")
public void consume(OrderEvent order) {
    // Automatically deserialized from JSON
    log.info("Received order: {}", order.getId());
}
```

Avro Serialization (Most Popular in Production)

Why Avro:

- ✓ Schema enforced – no bad data
- ✓ Compact binary format – smaller than JSON
- ✓ Schema evolution – add/remove fields safely
- ✓ Works with Schema Registry
- ✓ Fast serialization/deserialization

Maven dependency:

```
<dependency>
  <groupId>io.confluent</groupId>
  <artifactId>kafka-avro-serializer</artifactId>
  <version>7.5.0</version>
</dependency>
```

Define Avro schema (order.avsc):

```
{
  "type": "record",
  "name": "Order",
  "namespace": "com.myapp.avro",
  "fields": [
    {"name": "orderId", "type": "string"},
    {"name": "userId", "type": "string"},
    {"name": "amount", "type": "double"},
    {"name": "status", "type": "string"},
    {"name": "timestamp", "type": "long", "logicalType": "timestamp-millis"},
    {"name": "notes", "type": ["null", "string"], "default": null}
  ]
}
```

Producer with Avro:

```
Properties props = new Properties();
props.put("bootstrap.servers", "kafka-broker:9093");
props.put("security.protocol", "SSL");
props.put("ssl.truststore.location", "/certs/kafka.truststore.jks");
props.put("ssl.truststore.password", "yourpassword");

// Avro specific
props.put("key.serializer", StringSerializer.class.getName());
props.put("value.serializer", KafkaAvroSerializer.class.getName());
props.put("schema.registry.url", "https://schema-registry:8081");

// Schema registry SSL (if registry also uses certs)
props.put("schema.registry.ssl.truststore.location",
    "/certs/kafka.truststore.jks");
props.put("schema.registry.ssl.truststore.password", "yourpassword");

KafkaProducer<String, Order> producer = new KafkaProducer<>(props);

Order order = Order.newBuilder()
    .setOrderId("ORD-001")
    .setUserId("USR-123")
    .setAmount(99.99)
    .setStatus("CREATED")
    .setTimestamp(System.currentTimeMillis())
    .build();

producer.send(new ProducerRecord<>("orders", order.getId(), order));
```

Consumer with Avro:

```
props.put("key.deserializer", StringDeserializer.class.getName());
props.put("value.deserializer", KafkaAvroDeserializer.class.getName());
props.put("schema.registry.url", "https://schema-registry:8081");
props.put("specific.avro.reader", true); // Use generated class

KafkaConsumer<String, Order> consumer = new KafkaConsumer<>(props);
consumer.subscribe(List.of("orders"));

while (true) {
    ConsumerRecords<String, Order> records = consumer.poll(Duration.ofMillis(100));
    for (ConsumerRecord<String, Order> record : records) {
        Order order = record.value();
        System.out.println("Order: " + order.getId());
    }
}
```

Spring Boot with Avro:

```
spring:
  kafka:
    bootstrap-servers: kafka-broker:9093
    properties:
      security.protocol: SSL
      ssl.truststore.location: /certs/kafka.truststore.jks
      ssl.truststore.password: yourpassword
      schema.registry.url: https://schema-registry:8081
      specific.avro.reader: true
    producer:
      key-serializer: org.apache.kafka.common.serialization.StringSerializer
      value-serializer: io.confluent.kafka.serializers.KafkaAvroSerializer
    consumer:
      key-deserializer: org.apache.kafka.common.serialization.StringDeserializer
      value-deserializer: io.confluent.kafka.serializers.KafkaAvroDeserializer
      group-id: my-group
```

Protobuf Serialization

```
<dependency>
  <groupId>io.confluent</groupId>
  <artifactId>kafka-protobuf-serializer</artifactId>
  <version>7.5.0</version>
</dependency>
```

```
// order.proto
syntax = "proto3";
package com.myapp;

message Order {
  string order_id = 1;
  string user_id = 2;
  double amount = 3;
  string status = 4;
  int64 timestamp = 5;
}
```

```
props.put("value.serializer",
    KafkaProtobufSerializer.class.getName());
props.put("value.deserializer",
    KafkaProtobufDeserializer.class.getName());
props.put("schema.registry.url", "https://schema-registry:8081");
props.put("protobuf.value.deserializer.class",
    Order.class.getName());
```

Handling Deserialization Errors


```
// Spring Kafka – ErrorHandlingDeserializer wraps your deserializer
// Routes bad messages to DLT instead of crashing consumer
```

```
spring:
  kafka:
    consumer:
      value-deserializer: org.springframework.kafka.support.serializer.ErrorHandlingDeserializer
      properties:
        spring.deserializer.value.delegate.class: >
          io.confluent.kafka.serializers.KafkaAvroDeserializer
```

```
// Handle deserialization errors in listener
@KafkaListener(topics = "orders")
public void listen(ConsumerRecord<String, Object> record) {
  if (record.value() instanceof DeserializationException) {
    DeserializationException ex =
      (DeserializationException) record.value();
    log.error("Bad message at offset {}: {}",
      record.offset(), ex.getMessage());
    dltProducer.send("orders.DLT",
      record.key(), ex.getData()); // raw bytes to DLT
    return;
  }
  processMessage((Order) record.value());
}
```

PART 3: Kafka Configuration (Complete Reference)

Broker Configuration (server.properties)

```
#-----
# Core Broker Settings
#-----
broker.id=1
listeners=PLAINTEXT://0.0.0.0:9092,SSL://0.0.0.0:9093
advertised.listeners=PLAINTEXT://my-broker:9092,SSL://my-broker:9093
listener.security.protocol.map=PLAINTEXT:PLAINTEXT,SSL:SSL
inter.broker.listener.name=SSL

#-----
# Log / Storage
#-----
log.dirs=/kafka/data
num.partitions=3
default.replication.factor=3
min.insync.replicas=2
log.retention.hours=168          # 7 days
log.segment.bytes=1073741824    # 1GB per segment
log.retention.check.interval.ms=300000
```

```
auto.create.topics.enable=false # Disable in production
```

```
#-----
```

```
# Replication
```

```
#-----
```

```
unclean.leader.election.enable=false
```

```
auto.leader.rebalance.enable=true
```

```
leader.imbalance.check.interval.seconds=300
```

```
#-----
```

```
# Performance
```

```
#-----
```

```
num.network.threads=8
```

```
num.io.threads=16
```

```
socket.send.buffer.bytes=1048576
```

```
socket.receive.buffer.bytes=1048576
```

```
num.replica.fetchers=4
```

```
#-----
```

```
# SSL
```

```
#-----
```

```
ssl.keystore.location=/certs/kafka.keystore.jks
```

```
ssl.keystore.password=yourpassword
```

```
ssl.key.password=yourpassword
```

```
ssl.truststore.location=/certs/kafka.truststore.jks
```

```
ssl.truststore.password=yourpassword
```

```
ssl.client.auth=required
```

```
ssl.protocol=TLSv1.3
```

```
ssl.enabled.protocols=TLSv1.3,TLSv1.2
```

Producer Configuration (Complete)

```
#-----
```

```
# Connection
```

```
#-----
```

```
bootstrap.servers=kafka-broker:9093
```

```
security.protocol=SSL
```

```
ssl.truststore.location=/certs/kafka.truststore.jks
```

```
ssl.truststore.password=yourpassword
```

```
ssl.keystore.location=/certs/kafka.keystore.jks
```

```
ssl.keystore.password=yourpassword
```

```
#-----
```

```
# Serialization
```

```
#-----
```

```
key.serializer=org.apache.kafka.common.serialization.StringSerializer
```

```
value.serializer=io.confluent.kafka.serializers.KafkaAvroSerializer
```

```
schema.registry.url=https://schema-registry:8081
```

```
#-----
```

```
# Reliability
```

```
#-----
acks=all
enable.idempotence=true
retries=2147483647
max.in.flight.requests.per.connection=5
delivery.timeout.ms=120000

#-----
# Performance / Batching
#-----
linger.ms=10
batch.size=65536
buffer.memory=67108864
compression.type=snappy
max.block.ms=60000

#-----
# Transactions (if needed)
#-----
transactional.id=my-service-producer-1
transaction.timeout.ms=60000
```

Consumer Configuration (Complete)

```
#-----
# Connection
#-----
bootstrap.servers=kafka-broker:9093
security.protocol=SSL
ssl.truststore.location=/certs/kafka.truststore.jks
ssl.truststore.password=yourpassword
ssl.keystore.location=/certs/kafka.keystore.jks
ssl.keystore.password=yourpassword

#-----
# Deserialization
#-----
key.deserializer=org.apache.kafka.common.serialization.StringDeserializer
value.deserializer=io.confluent.kafka.serializers.KafkaAvroDeserializer
schema.registry.url=https://schema-registry:8081
specific.avro.reader=true

#-----
# Consumer Group
#-----
group.id=my-consumer-group
group.instance.id=my-consumer-1 # static membership
auto.offset.reset=earliest
enable.auto.commit=false

#-----
```

```
# Performance
#-----
fetch.min.bytes=1024
fetch.max.wait.ms=500
max.poll.records=500
max.poll.interval.ms=300000

#-----
# Session / Heartbeat
#-----
session.timeout.ms=45000
heartbeat.interval.ms=10000

#-----
# Transactions (exactly-once)
#-----
isolation.level=read_committed
```

Spring Boot Full Configuration

```
spring:
  kafka:
    bootstrap-servers: kafka-broker:9093

    # SSL Certificates
    properties:
      security.protocol: SSL
      ssl.truststore.location: /certs/kafka.truststore.jks
      ssl.truststore.password: ${KAFKA_TRUSTSTORE_PASSWORD}
      ssl.keystore.location: /certs/kafka.keystore.jks
      ssl.keystore.password: ${KAFKA_KEYSTORE_PASSWORD}
      ssl.key.password: ${KAFKA_KEY_PASSWORD}
      ssl.protocol: TLSv1.3
      schema.registry.url: https://schema-registry:8081
      schema.registry.ssl.truststore.location: /certs/kafka.truststore.jks
      schema.registry.ssl.truststore.password: ${KAFKA_TRUSTSTORE_PASSWORD}

    producer:
      key-serializer: org.apache.kafka.common.serialization.StringSerializer
      value-serializer: io.confluent.kafka.serializers.KafkaAvroSerializer
      acks: all
      retries: 2147483647
      properties:
        enable.idempotence: true
        linger.ms: 10
        batch.size: 65536
        compression.type: snappy

    consumer:
      key-deserializer: org.apache.kafka.common.serialization.StringDeserializer
```

```

value-deserializer: io.confluent.kafka.serializers.KafkaAvroDeserializer
group-id: my-service-group
auto-offset-reset: earliest
enable-auto-commit: false
max-poll-records: 500
properties:
  specific.avro.reader: true
  isolation.level: read_committed
  max.poll.interval.ms: 300000
  session.timeout.ms: 45000
  heartbeat.interval.ms: 10000

listener:
  ack-mode: manual
  concurrency: 3           # 3 listener threads
  type: batch              # batch or single

```

Externalizing Secrets (Never hardcode passwords)

```

# Use environment variables
ssl.truststore.password: ${KAFKA_TRUSTSTORE_PASSWORD}

```

```

# Or use Spring Cloud Config / Vault
spring:
  cloud:
    vault:
      uri: https://vault:8200
      authentication: KUBERNETES

```

```

// Or inject via @Value
@Value("${kafka.ssl.truststore.password}")
private String truststorePassword;

@Bean
public KafkaTemplate<String, Object> kafkaTemplate() {
    Map<String, Object> props = new HashMap<>();
    props.put("ssl.truststore.password", truststorePassword);
    // ...
}

```

Quick Troubleshooting Cheatsheet

Error	Cause	Fix
<code>SSLHandshakeException</code>	Wrong cert or expired	Check cert validity, verify truststore
<code>CertificateExpiredException</code>	Cert expired	Renew certificate

Error	Cause	Fix
<code>SerializationException</code>	Wrong serializer	Match serializer to data format
<code>UnknownTopicOrPartitionException</code>	Topic doesn't exist	Create topic or enable auto-create
<code>AuthorizationException</code>	No ACL permission	Grant ACLs to client principal
<code>SaslAuthenticationException</code>	Wrong credentials	Verify JAAS config username/password
<code>SchemaRegistryException</code>	Schema not found	Register schema or check registry URL
<code>NotEnoughReplicasException</code>	ISR < min.insync	Bring brokers back or lower min.insync

Want me to go deeper into any specific area — like setting up Schema Registry with SSL, Kubernetes Kafka config with cert secrets, or a specific language like Python or Node.js?

5/2/2026, 11:20:45 PM

👤 Did we cover most interview related to Kafka or anything pending

5/2/2026, 11:21:06 PM

Kafka Interview Coverage Review

What We've Covered

Area	Topics Covered
Core Concepts	Topics, partitions, offsets, brokers, ISR
Architecture	Leader/follower, controller, ZooKeeper, KRaft
Producers	acks, idempotence, transactions, batching
Consumers	Consumer groups, rebalancing, lag, offsets
CLI Operations	Create/delete topics, consume, produce, reset offsets
Monitoring	Lag monitoring, JMX metrics, Prometheus/Grafana
Security	SSL/TLS, SASL, ACLs, certificates
Serialization	JSON, Avro, Protobuf, error handling
Configuration	Broker, producer, consumer, Spring Boot
Kafka Streams	Joins, windowing, state stores
Kafka Connect	Source/sink, REST API, troubleshooting
Design Patterns	Event sourcing, CQRS, Saga, Outbox
Scenarios	Data loss, poison pill, rebalancing, spike handling
Performance	Tuning, OS settings, capacity planning
Testing	EmbeddedKafka, Testcontainers, TestDriver
Multi-DC	MirrorMaker2, active-active, active-passive

What's Still Pending

1. Kafka vs Other Systems (Very Common Interview Question)

Kafka vs RabbitMQ vs ActiveMQ vs Pulsar:

Feature	Kafka	RabbitMQ	ActiveMQ	Pulsar
Model	Log-based pub/sub	Message queue	Message queue	Log-based pub/sub
Retention	Configurable	Deleted after ack	Deleted after ack	Configurable
Throughput	Very high (millions/sec)	Moderate	Moderate	Very high
Ordering	Per partition	Per queue	Per queue	Per partition
Replay	Yes	No	No	Yes

Feature	Kafka	RabbitMQ	ActiveMQ	Pulsar
Multi-tenancy	Limited	Limited	Limited	Built-in
Geo-replication	MirrorMaker2	Federation	Network of brokers	Built-in
Best for	Streaming, event log	Task queues, RPC	Enterprise messaging	Cloud-native streaming

2. Kafka Guarantees & Edge Cases

Q: Can Kafka guarantee strict global ordering?

No – Kafka **only** guarantees ordering **WITHIN** a partition.

For **global** ordering:

- Use single **partition** (kills parallelism)
- Use application-level sequencing (sequence numbers **in** payload)
- Use Kafka Streams event-**time** processing **with** watermarks

Q: What happens if a producer sends a message and never gets an ack?

With retries enabled:

- **Producer retries automatically**
- **With `enable.idempotence=true` → no duplicates on retry**
- **Without idempotence → possible duplicate messages**

Without retries:

- **Producer throws `TimeoutException`**
- **Message may or may not have been written**
- **Application must handle this case**

Q: Can a consumer read from a follower replica?

Yes – since Kafka **2.4** **with** KIP-392

Useful **for**:

- Rack-aware **consumption** (read **from** same AZ)
- Reducing cross-AZ network costs

Config:

```
replica.selector.class=org.apache.kafka.common.replica.RackAwareReplicaSelector
```

Consumer config:

```
client.rack=us-east-1a # match broker rack
```

Q: What happens when log compaction runs?

Before compaction:

Key=user1 → **value=A** (**offset 0**)

Key=user2 → **value=B** (**offset 1**)

Key=user1 → **value=C** (**offset 2**)

Key=user1 → value=D (offset 3)

After compaction:

Key=user2 → value=B (offset 1) ← kept (only record)

Key=user1 → value=D (offset 3) ← kept (latest)

Offsets 0 and 2 are removed

3. Kafka Header Usage

Very common in microservices — rarely covered:

```
// Producer — add metadata headers
ProducerRecord<String, String> record = new ProducerRecord<>(
    "orders", "key1", "payload");

record.headers()
    .add("correlationId", UUID.randomUUID().toString().getBytes())
    .add("source-service", "order-service".getBytes())
    .add("event-type", "ORDER_CREATED".getBytes())
    .add("version", "v2".getBytes());

producer.send(record);

// Consumer — read headers
@KafkaListener(topics = "orders")
public void listen(ConsumerRecord<String, String> record) {
    String correlationId = new String(
        record.headers().lastHeader("correlationId").value());
    String eventType = new String(
        record.headers().lastHeader("event-type").value());

    MDC.put("correlationId", correlationId); // Add to logs
    processMessage(record.value(), eventType);
}
```

Common header use cases:

correlationId	→ distributed tracing across services
event-type	→ route to different handlers without deserializing
version	→ handle multiple schema versions
source-service	→ know which service produced the message
retry-count	→ track how many times message has been retried
dead-letter-reason	→ why message was sent to DLT

4. Kafka Compacted Topics — Practical Use

Q: When would you use a compacted topic in real work?

Use cases:

- User profile store (latest profile per userId)
- Product catalog (latest product info per productId)
- Configuration store (latest config per key)
- CDC (Change Data Capture) – latest DB row state
- Session store for Kafka Streams stateful apps

```
# Create compacted topic
kafka-topics.sh --create \
  --bootstrap-server localhost:9092 \
  --topic user-profiles \
  --partitions 6 \
  --replication-factor 3 \
  --config cleanup.policy=compact \
  --config min.cleanable.dirty.ratio=0.1 \
  --config segment.ms=3600000

# Delete a key (tombstone message – null value)
kafka-console-producer.sh \
  --bootstrap-server localhost:9092 \
  --topic user-profiles \
  --property "key.separator=:" \
  --property "parse.key=true" \
  --property "null.marker=NULL"
# Type: user-123:NULL → deletes user-123 key
```

5. Kafka Quotas & Multi-Tenancy

Q: How do you prevent one service from flooding Kafka?

```
# Per-client producer quota
kafka-configs.sh --bootstrap-server localhost:9092 \
  --alter --entity-type clients \
  --entity-name analytics-service \
  --add-config producer_byte_rate=5242880 # 5MB/s

# Per-user quota
kafka-configs.sh --bootstrap-server localhost:9092 \
  --alter --entity-type users \
  --entity-name service-account-x \
  --add-config producer_byte_rate=10485760,consumer_byte_rate=10485760

# Per-IP quota
kafka-configs.sh --bootstrap-server localhost:9092 \
  --alter --entity-type ips \
```

```
--entity-name 10.0.0.5 \  
--add-config producer_byte_rate=1048576
```

When quota is exceeded:

```
Producer gets throttled → delayed responses  
Broker metric: kafka_server_produce_throttle_time_avg  
Alert if throttle time > 0 consistently
```

6. Kafka Rebalance Strategies Deep Dive

Q: Explain all partition assignment strategies:

```
RANGE (default):  
  Partitions sorted numerically, divided by consumer count  
  Consumer 0 → P0, P1, P2  
  Consumer 1 → P3, P4, P5  
  Problem: first consumer always gets more if uneven  
  
ROUND_ROBIN:  
  Partitions distributed evenly one by one  
  Consumer 0 → P0, P2, P4  
  Consumer 1 → P1, P3, P5  
  Better balance but all partitions move on rebalance  
  
STICKY (recommended):  
  Like RoundRobin but minimizes partition movement  
  On rebalance – keeps existing assignments where possible  
  Best for reducing rebalance impact  
  
COOPERATIVE_STICKY (best for production):  
  Incremental rebalance – only moves partitions that need to move  
  Consumer keeps processing during rebalance  
  No stop-the-world rebalance pause
```

```
# Set in consumer config  
partition.assignment.strategy=\n  org.apache.kafka.clients.consumer.CooperativeStickyAssignor
```

7. Kafka Upgrade & Maintenance

Q: How do you do a rolling upgrade of Kafka brokers?

1. Update one broker at a time
2. Stop broker → upgrade → restart
3. Wait for ISR to recover before next broker
4. Monitor under-replicated partitions during upgrade

```
# Check ISR recovering
kafka-topics.sh --bootstrap-server localhost:9092 \
  --describe --under-replicated-partitions
# Should return empty before proceeding to next broker

# Set inter.broker.protocol.version during upgrade
# Keep old version until all brokers upgraded
inter.broker.protocol.version=3.4 # old version
# After all brokers upgraded, bump to new version
inter.broker.protocol.version=3.6
```

Q: How do you rebalance partition leadership across brokers?

```
# Auto rebalance preferred leaders
kafka-leader-election.sh \
  --bootstrap-server localhost:9092 \
  --election-type PREFERRED \
  --all-topic-partitions

# Manual partition reassignment
kafka-reassign-partitions.sh \
  --bootstrap-server localhost:9092 \
  --reassignment-json-file reassignment.json \
  --execute

# Check reassignment progress
kafka-reassign-partitions.sh \
  --bootstrap-server localhost:9092 \
  --reassignment-json-file reassignment.json \
  --verify
```

8. Dead Letter Queue (DLQ) Patterns

Complete DLQ implementation — very practical:

```
@Configuration
public class KafkaConfig {

    @Bean
    public DefaultErrorHandler errorHandler(
        KafkaTemplate<String, Object> template) {

        // Retry 3 times with exponential backoff
        ExponentialBackOffWithMaxRetries backoff =
            new ExponentialBackOffWithMaxRetries(3);
        backoff.setInitialInterval(1000L); // 1 second
        backoff.setMultiplier(2.0); // 2x each retry
        backoff.setMaxInterval(10000L); // max 10 seconds
```

```

// After retries exhausted → send to DLT
DeadLetterPublishingRecoverer recoverer =
    new DeadLetterPublishingRecoverer(template,
        (record, ex) -> new TopicPartition(
            record.topic() + ".DLT",
            record.partition()));

    return new DefaultErrorHandler(recoverer, backoff);
}
}

// Monitor DLT messages
@KafkaListener(topics = "orders.DLT")
public void handleDLT(
    ConsumerRecord<String, Object> record,
    @Header(KafkaHeaders.EXCEPTION_MESSAGE) String errorMsg) {
    log.error("DLT message from topic={} offset={} error={}",
        record.topic(), record.offset(), errorMsg);
    // Alert, store in DB, notify team
    alertingService.sendAlert(record, errorMsg);
}

```

9. Kafka in Kubernetes

Q: How do you deploy Kafka on Kubernetes?

```

# Using Strimzi Operator (most popular)
apiVersion: kafka.strimzi.io/v1beta2
kind: Kafka
metadata:
  name: my-cluster
spec:
  kafka:
    version: 3.6.0
    replicas: 3
    listeners:
      - name: plain
        port: 9092
        type: internal
        tls: false
      - name: tls
        port: 9093
        type: internal
        tls: true
        authentication:
          type: tls
  config:
    offsets.topic.replication.factor: 3
    transaction.state.log.replication.factor: 3

```

```

    min.insync.replicas: 2
    default.replication.factor: 3
  storage:
    type: jbod
    volumes:
      - id: 0
        type: persistent-claim
        size: 500Gi
        class: fast-ssd
  resources:
    requests:
      memory: 8Gi
      cpu: "2"
    limits:
      memory: 16Gi
      cpu: "4"
  zookeeper:
    replicas: 3
    storage:
      type: persistent-claim
      size: 10Gi

```

Mounting SSL certs as Kubernetes secrets:

```

# Create secret from cert files
kubectl create secret generic kafka-certs \
  --from-file=truststore.jks=kafka.truststore.jks \
  --from-file=keystore.jks=kafka.keystore.jks

# Mount in pod
spec:
  containers:
    - name: my-service
      volumeMounts:
        - name: kafka-certs
          mountPath: /certs
          readOnly: true
  volumes:
    - name: kafka-certs
      secret:
        secretName: kafka-certs

```

10. Common Kafka Interview Coding Questions

Q: Write a producer that guarantees exactly-once delivery:

```

public class ExactlyOnceProducer {
    private final KafkaProducer<String, String> producer;

```

```

public ExactlyOnceProducer(String bootstrapServers) {
    Properties props = new Properties();
    props.put("bootstrap.servers", bootstrapServers);
    props.put("key.serializer", StringSerializer.class.getName());
    props.put("value.serializer", StringSerializer.class.getName());
    props.put("enable.idempotence", "true");
    props.put("acks", "all");
    props.put("retries", Integer.MAX_VALUE);
    props.put("transactional.id", "my-tx-producer-1");

    this.producer = new KafkaProducer<>(props);
    producer.initTransactions();
}

public void sendWithTransaction(String topic,
                                String key, String value) {
    producer.beginTransaction();
    try {
        producer.send(new ProducerRecord<>(topic, key, value));
        producer.commitTransaction();
    } catch (Exception e) {
        producer.abortTransaction();
        throw new RuntimeException("Transaction failed", e);
    }
}
}

```

Q: Write a consumer with manual offset management:

```

public class ManualOffsetConsumer {
    public void consume(String topic, String groupId) {
        Properties props = new Properties();
        props.put("bootstrap.servers", "localhost:9092");
        props.put("group.id", groupId);
        props.put("enable.auto.commit", "false");
        props.put("key.deserializer", StringDeserializer.class.getName());
        props.put("value.deserializer", StringDeserializer.class.getName());

        KafkaConsumer<String, String> consumer = new KafkaConsumer<>(props);
        consumer.subscribe(List.of(topic));

        try {
            while (true) {
                ConsumerRecords<String, String> records =
                    consumer.poll(Duration.ofMillis(100));

                for (ConsumerRecord<String, String> record : records) {
                    try {
                        processMessage(record);
                        // Commit per partition after success
                        consumer.commitSync(Collections.singletonMap(
                            new TopicPartition(

```

```

        record.topic(), record.partition()),
        new OffsetAndMetadata(record.offset() + 1)
    ));
} catch (Exception e) {
    log.error("Failed at offset {}", record.offset());
    // Don't commit – message will be reprocessed
}
}
}
} finally {
    consumer.close();
}
}
}
}

```

Complete Coverage Checklist

Topic	Status
Core concepts	✓ Covered
Architecture internals	✓ Covered
Producer/Consumer configs	✓ Covered
CLI operations	✓ Covered
SSL/Certificates	✓ Covered
Serialization (JSON/Avro/Protobuf)	✓ Covered
Schema Registry	✓ Covered
Kafka Streams	✓ Covered
Kafka Connect	✓ Covered
Monitoring & alerting	✓ Covered
Scenario-based debugging	✓ Covered
Design patterns	✓ Covered
Multi-datacenter	✓ Covered
Testing	✓ Covered
Kafka vs other systems	✓ Now covered
Kafka headers	✓ Now covered
Compacted topics practical	✓ Now covered
Quotas & multi-tenancy	✓ Now covered
Rebalance strategies deep dive	✓ Now covered
Rolling upgrades	✓ Now covered
DLQ patterns complete	✓ Now covered

Topic	Status
Kafka on Kubernetes	✓ Now covered
Coding questions	✓ Now covered
KRaft internals	✓ Now covered
Tiered storage	✓ Now covered
Performance tuning	✓ Now covered

Final Tips for Kafka Interviews

Junior level → Focus on core concepts, CLI, basic producer/consumer
 Mid level → Focus on configs, debugging, serialization, connect
 Senior level → Focus on design patterns, exactly-once, performance, multi-DC, Kubernetes, schema evolution
 Architect level → Focus on capacity planning, security, tiered storage, Kafka vs alternatives, stream processing design

You are now comprehensively covered for Kafka interviews at all levels. Would you like me to do a **mock interview** with rapid-fire Kafka questions to test your knowledge?

5/2/2026, 11:22:11 PM

What is throughput here When rebalancing happens Kafka must be doing slow processing right

5/3/2026, 7:16:05 AM

Great questions! Let me explain both clearly.

1. What is Throughput in Kafka?

Throughput = how much data Kafka can process per second

Throughput = Messages per second (or) MB/s (or) GB/s

Simple Analogy

Think of a highway:

Low throughput → single lane road → few cars per minute

High throughput → 10-lane highway → thousands of cars per minute

Kafka is the 10-lane highway for data

Practical Numbers

Single Kafka partition → ~10-100 MB/s

Single Kafka broker → ~100-500 MB/s

Kafka cluster (3 nodes) → ~1-3 GB/s

LinkedIn's Kafka → ~7 trillion messages/day

What Affects Throughput?

Higher throughput ↑	Lower throughput ↓
More partitions	Single partition
Compression enabled	No compression
Large batch size (linger.ms)	Send one message at a time
Sequential disk writes	Random writes
More brokers	Single broker
Fast network	Slow network
acks=1	acks=all (slightly slower)

Throughput vs Latency Tradeoff

High throughput setting:

linger.ms=20 → wait 20ms to fill a batch

batch.size=65536 → send 64KB at once

Result: MORE data per second but HIGHER delay per message

Low latency setting:

`linger.ms=0` → send immediately

`batch.size=16384` → small batches

Result: FASTER **per** message but LESS data **per second**

Choose based **on** your use **case**:

Trading systems → low latency (speed matters)

Log aggregation → high throughput (volume matters)

Payment events → balance **both**

How to Measure Throughput

```
# Built-in Kafka throughput test tool
kafka-producer-perf-test.sh \
  --topic perf-test \
  --num-records 1000000 \
  --record-size 1024 \
  --throughput -1 \
  --producer-props bootstrap.servers=localhost:9092

# Output looks like:
# 500000 records sent, 98500 records/sec (96.2 MB/sec),
# 12.3 ms avg latency, 245 ms max latency

# Consumer throughput test
kafka-consumer-perf-test.sh \
  --bootstrap-server localhost:9092 \
  --topic perf-test \
  --messages 1000000
```

2. Does Rebalancing Cause Slow Processing?

Yes — rebalancing causes a "stop the world" pause. Here's exactly what happens:

What Happens During Rebalance

```
Normal state:
Consumer A → reads Partition 0, 1
Consumer B → reads Partition 2, 3
Consumer C → reads Partition 4, 5
      ↓
Trigger (new consumer joins / consumer dies)
      ↓
REBALANCE STARTS
_____
ALL consumers stop polling    ← processing PAUSES here
ALL partition assignments dropped
```

Group coordinator recalculates assignments
ALL consumers get new assignments

↓

Processing resumes
Lag builds up during pause

How Long Does a Rebalance Take?

Typical rebalance duration:

Small group (3 consumers) → 1-5 seconds pause
Medium group (20 consumers) → 5-30 seconds pause
Large group (100 consumers) → 30-120 seconds pause

During this time:

- No messages consumed
- Producer keeps writing
- Consumer lag grows
- Downstream systems may time out

What Triggers Rebalancing?

1. New consumer joins the group
2. Consumer leaves or crashes
3. Consumer heartbeat times out
4. Consumer takes too long to process (max.poll.interval.ms exceeded)
5. New partitions added to topic
6. Subscription change
7. Application deployment / restart

How to Minimize Rebalance Impact

Fix 1 — Cooperative Sticky Assignor (biggest improvement):

Old way → stop everything, reassign all partitions
New way → only move partitions that need to move

```
partition.assignment.strategy=\n  org.apache.kafka.clients.consumer.CooperativeStickyAssignor
```

Without Cooperative:		With Cooperative:
All 6 partitions revoked	→	Only 2 partitions moved
All consumers pause	→	Other 4 keep processing
Full stop-the-world	→	Incremental, minimal pause

Fix 2 — Static Group Membership (eliminates rebalance on restart):

```
group.instance.id=consumer-pod-1 # unique per instance\nsession.timeout.ms=300000        # 5 min — wait before rebalancing
```

Without `static` membership:

Pod restarts → leaves `group` → REBALANCE → rejoins → REBALANCE

With `static` membership:

Pod restarts → rejoins `with` same ID → NO rebalance

(`as long as` it comes back within `session.timeout.ms`)

Fix 3 — Tune poll and heartbeat timeouts:

Give consumers more time to process before being kicked out

`max.poll.interval.ms=600000` # 10 minutes

`session.timeout.ms=60000` # 1 minute

`heartbeat.interval.ms=20000` # every 20 seconds

Reduce records per poll if processing is slow

`max.poll.records=100` # process less per batch

Rebalance Types Summary

EAGER rebalance (`old default`):

- `All` partitions revoked `from` everyone
- `Full` stop, reassign everything
- Simple but very disruptive

COOPERATIVE rebalance (recommended):

- `Only` affected partitions move
- Others keep processing
- Two rounds but much less impact
- Use `CooperativeStickyAssignor`

Quick Summary

Throughput = how much data flows through Kafka per second
→ make it higher `with` batching, compression, partitions

Rebalancing = yes it pauses processing (`stop-the-world`)
→ minimize `with` `CooperativeStickyAssignor`
→ eliminate `on` restarts `with` `Static Group Membership`
→ tune timeouts `to` avoid unnecessary rebalances

Want me to go deeper into either topic — like how to benchmark throughput for your specific use case, or how Cooperative rebalancing works internally step by step?

5/3/2026, 7:16:38 AM

 Go deeper on these topics

performance tuning, or system design with Kafka?

5/3/2026, 7:24:07 AM

Kafka Performance Tuning & System Design — Deep Dive

PART 1: Performance Tuning

The Performance Tuning Mental Model

Performance problems fall **into 3** buckets:

1. **THROUGHPUT** problems → **not** enough data flowing through
Symptoms: producer backpressure, growing lag, slow consumers
2. **LATENCY** problems → messages taking too long **end to end**
Symptoms: delays **between** produce **and** consume
3. **STABILITY** problems → Kafka works but keeps **having** issues
Symptoms: rebalances, broker restarts, OOM errors

Broker-Level Tuning

JVM Tuning

```
# Kafka startup script – set heap size
export KAFKA_HEAP_OPTS="-Xms6g -Xmx6g"
# Rule: 4-6GB heap for brokers. Never more than 6GB
# More heap → longer GC pauses → higher latency spikes

# Use G1GC – best for Kafka
export KAFKA_JVM_PERFORMANCE_OPTS="
  -server
  -XX:+UseG1GC
  -XX:MaxGCPauseMillis=20
  -XX:InitiatingHeapOccupancyPercent=35
  -XX:+ExplicitGCInvokesConcurrent
  -XX:MaxInlineLevel=15
  -Djava.awt.headless=true"
```

Why limit heap to 6GB?

Kafka heavily uses OS page cache – NOT JVM heap

Page cache = RAM outside JVM heap

32GB RAM broker:

JVM heap → 6GB (Kafka process)

OS page cache → 26GB (hot data cached here)

If you give JVM 20GB heap:

JVM heap → 20GB

OS page cache → 12GB (less cache = more disk reads = slower)

OS-Level Tuning

```
# 1. File descriptors – Kafka opens many files
echo "* hard nfile 100000" >> /etc/security/limits.conf
echo "* soft nfile 100000" >> /etc/security/limits.conf
ulimit -n 100000

# 2. Virtual memory – avoid swapping
sysctl -w vm.swappiness=1          # almost never swap
sysctl -w vm.dirty_background_ratio=5 # start flushing at 5% dirty
sysctl -w vm.dirty_ratio=80       # hard limit at 80%

# 3. Network buffers – high throughput
sysctl -w net.core.rmem_default=134217728
sysctl -w net.core.wmem_default=134217728
sysctl -w net.core.rmem_max=134217728
sysctl -w net.core.wmem_max=134217728
sysctl -w net.ipv4.tcp_rmem="4096 87380 134217728"
sysctl -w net.ipv4.tcp_wmem="4096 65536 134217728"

# 4. Use XFS filesystem for Kafka log dirs
# XFS handles many small files better than ext4
mkfs.xfs -f /dev/sdb
mount -o noatime,nodiratime /dev/sdb /kafka/data
# noatime = don't update file access time on read (huge improvement)

# 5. Disk scheduler – use deadline or noop for SSDs
echo deadline > /sys/block/sdb/queue/scheduler
# For NVMe SSDs use noop
echo noop > /sys/block/nvme0n1/queue/scheduler
```

Broker Config Tuning

```
#-----
# Network & IO threads
#-----
# Rule: num.network.threads = number of CPU cores / 2
num.network.threads=8

# Rule: num.io.threads = number of disks x 2
```

```
num.io.threads=16

# Queue size for network requests
queued.max.requests=1000

#-----
# Log performance
#-----
# Don't flush to disk manually – let OS handle it
# OS page cache is much faster
log.flush.interval.messages=Long.MAX_VALUE
log.flush.interval.ms=Long.MAX_VALUE

# Large segment size = fewer segment rolls = better performance
log.segment.bytes=1073741824      # 1GB

# How often to check for log cleanup
log.retention.check.interval.ms=300000

#-----
# Replication performance
#-----
# More fetcher threads = faster replication
num.replica.fetchers=4

# Large fetch size = fewer round trips
replica.fetch.max.bytes=10485760   # 10MB
replica.fetch.min.bytes=1
replica.fetch.wait.max.ms=500

#-----
# Socket buffers
#-----
socket.send.buffer.bytes=1048576   # 1MB
socket.receive.buffer.bytes=1048576 # 1MB
socket.request.max.bytes=104857600  # 100MB max request size
```

Producer Tuning — Deep Dive

Batching & Compression (Biggest Wins)

Without batching:

```
Message 1 → network call
Message 2 → network call
Message 3 → network call
3 network round trips = slow
```

With batching (linger.ms=20):


```
Wait 20ms → collect 1000 messages → 1 network call
1 network round trip = fast
```

Throughput-optimized producer

```
bootstrap.servers=kafka-broker:9092
```

```
acks=1 # faster than acks=all for non-critical
```

```
compression.type=lz4 # best balance of speed vs ratio
```

```
linger.ms=20 # wait 20ms to fill batches
```

```
batch.size=131072 # 128KB batch size
```

```
buffer.memory=134217728 # 128MB total buffer
```

```
max.in.flight.requests.per.connection=5
```

Latency-optimized producer

```
acks=1
```

```
compression.type=none # no compression overhead
```

```
linger.ms=0 # send immediately
```

```
batch.size=16384 # small batches
```

```
max.block.ms=5000
```

Compression Comparison

Algorithm	CPU usage	Compression ratio	Speed
none	0%	1x	fastest send
gzip	high	4-5x	slow CPU
snappy	medium	2-3x	fast
lz4	low	2-3x	very fast ← recommended
zstd	medium	4-5x	fast ← best ratio+speed

Rule of thumb:

High CPU available → zstd (best ratio)

Low CPU / high speed → lz4 (best balance)

Network is bottleneck → gzip (highest ratio)

Producer Performance Test

```
// Benchmark your producer
```

```
public class ProducerBenchmark {
```

```
    public static void main(String[] args) throws Exception {
```

```
        Properties props = new Properties();
```

```
        props.put("bootstrap.servers", "localhost:9092");
```

```
        props.put("key.serializer", StringSerializer.class.getName());
```

```
        props.put("value.serializer", StringSerializer.class.getName());
```

```
        props.put("linger.ms", "20");
```

```
        props.put("batch.size", "131072");
```

```
        props.put("compression.type", "lz4");
```

```
        KafkaProducer<String, String> producer = new KafkaProducer<>(props);
```

```
        String payload = "x".repeat(1024); // 1KB message
```

```
        long start = System.currentTimeMillis();
```

```
        int count = 1_000_000;
```

```

        for (int i = 0; i < count; i++) {
            producer.send(new ProducerRecord<>("perf-test",
                String.valueOf(i), payload));
        }
        producer.flush();

        long elapsed = System.currentTimeMillis() - start;
        System.out.printf("Sent %d messages in %dms\n", count, elapsed);
        System.out.printf("Throughput: %.0f msg/sec, %.1f MB/sec\n",
            count * 1000.0 / elapsed,
            count * 1024.0 / elapsed / 1024);

        producer.close();
    }
}

```

Consumer Tuning — Deep Dive

Fetch Tuning

Throughput-optimized consumer

```

fetch.min.bytes=50000          # wait until 50KB available
fetch.max.wait.ms=500          # or 500ms – whichever first
max.partition.fetch.bytes=1048576 # 1MB per partition per fetch
max.poll.records=1000          # process 1000 records per poll

```

Latency-optimized consumer

```

fetch.min.bytes=1              # return immediately even 1 byte
fetch.max.wait.ms=10           # max 10ms wait
max.poll.records=100           # smaller batches

```

Parallel Processing Within Consumer

```

// Problem: Single-threaded consumer is slow
// Solution: Process records in parallel using thread pool

@KafkaListener(topics = "orders")
public void listen(List<ConsumerRecord<String, String>> records,
    Acknowledgment ack) {

    // Split records into parallel tasks
    ExecutorService executor = Executors.newFixedThreadPool(10);
    List<Future<?>> futures = new ArrayList<>();

    for (ConsumerRecord<String, String> record : records) {
        futures.add(executor.submit(() -> {
            processRecord(record); // parallel processing
        }));
    }
}

```

```

// Wait for all to complete
for (Future<?> future : futures) {
    future.get(); // throws if any task failed
}

ack.acknowledge(); // commit after all processed
}

```

Consumer Lag Monitoring & Autoscaling

```

# Kubernetes HPA based on Kafka lag
# Using KEDA (Kubernetes Event Driven Autoscaler)
apiVersion: keda.sh/v1alpha1
kind: ScaledObject
metadata:
  name: consumer-scaler
spec:
  scaleTargetRef:
    name: my-consumer-deployment
  minReplicaCount: 2
  maxReplicaCount: 10          # never exceed partition count
  triggers:
    - type: kafka
      metadata:
        bootstrapServers: kafka-broker:9092
        consumerGroup: my-group
        topic: orders
        lagThreshold: "1000"   # scale up if lag > 1000
        offsetResetPolicy: latest

```

Full Performance Tuning Checklist

BROKER:

- ☐ JVM heap 4-6GB with G1GC
- ☐ XFS filesystem with noatime mount
- ☐ vm.swappiness=1
- ☐ num.io.threads = 2x disk count
- ☐ Disable log.flush (let OS handle)
- ☐ Separate disks for logs vs OS

PRODUCER:

- ☐ compression.type=lz4 or zstd
- ☐ linger.ms=5-20 (throughput) or 0 (latency)
- ☐ batch.size=65536-131072
- ☐ acks=all + enable.idempotence=true (reliability)
- ☐ Monitor record-error-rate metric

CONSUMER:

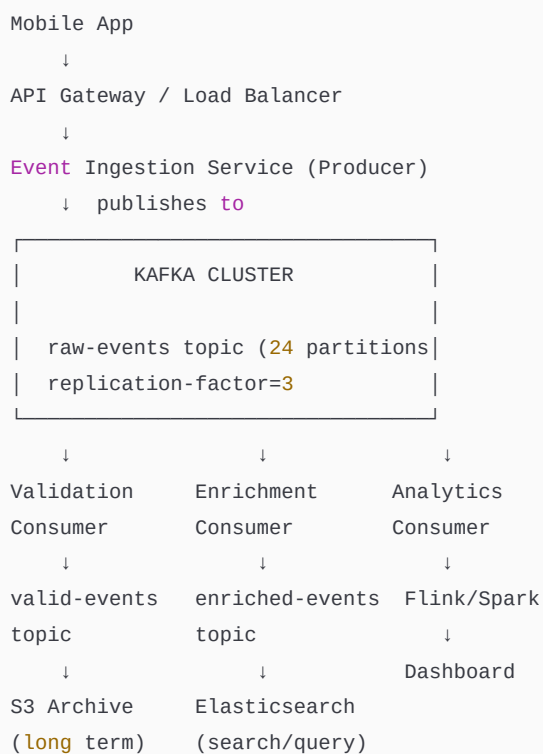
- ❑ `fetch.min.bytes > 1` for throughput
- ❑ `max.poll.records` tuned to processing speed
- ❑ Parallel processing within consumer
- ❑ `CooperativeStickyAssignor` for rebalancing
- ❑ Monitor consumer lag continuously

PART 2: System Design with Kafka

Design 1 — Real-Time Event Streaming Platform

Requirement: Process 10 million user events per day from a mobile app

Architecture:



Capacity calculation:

10M events/day
= 10,000,000 / 86,400 seconds
= ~116 events/second average
= ~500 events/second peak (4x average)

Each event = 2KB average
Peak throughput = 500 x 2KB = 1MB/s

Partitions needed:

1 partition handles ~50MB/s safely
We need 1MB/s → 1 partition is enough
BUT **for** parallelism → use 24 partitions

Retention:

500 events/s x 2KB x 86400s x 7 days = ~600GB
With 3x replication = ~1.8TB total storage

Topic design:

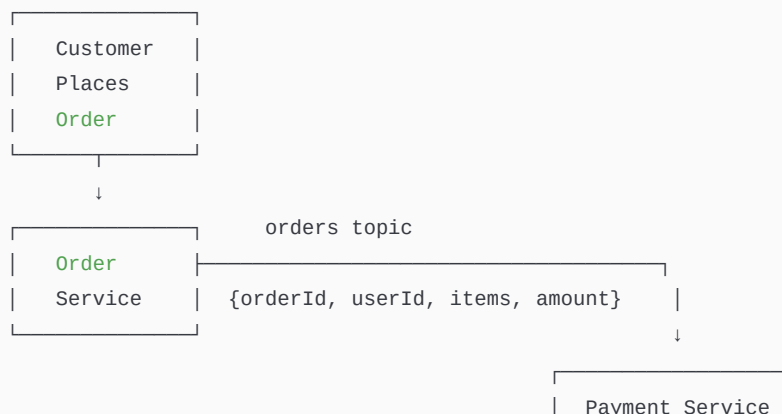
```
# Raw events – high volume, short retention
kafka-topics.sh --create \
  --topic raw-events \
  --partitions 24 \
  --replication-factor 3 \
  --config retention.ms=86400000 \      # 1 day
  --config compression.type=lz4

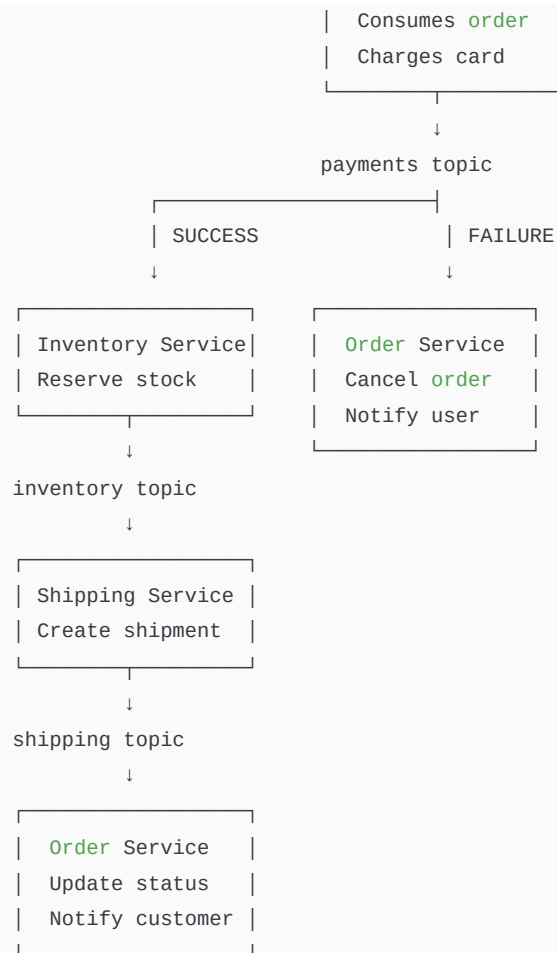
# Validated events – medium volume, longer retention
kafka-topics.sh --create \
  --topic valid-events \
  --partitions 12 \
  --replication-factor 3 \
  --config retention.ms=604800000      # 7 days

# Dead letter – failed events
kafka-topics.sh --create \
  --topic raw-events.DLT \
  --partitions 6 \
  --replication-factor 3 \
  --config retention.ms=2592000000     # 30 days
```

Design 2 — Order Processing System (Saga Pattern)

Requirement: E-commerce order flow with multiple microservices





Key design decisions:

1. Use orderId as partition key
 - All events for same order go to same partition
 - Guaranteed ordering per order
2. Each service has its own consumer group
 - Services are independent
 - One slow service doesn't affect others
3. Compensating transactions on failure
 - Payment fails → publish to payment-failed topic
 - Order service consumes → cancels order
 - No distributed transaction needed
4. Outbox pattern for each service
 - DB write + outbox write in same transaction
 - Debezium reads outbox → publishes to Kafka
 - Prevents dual-write inconsistency

Topic structure:

- | | |
|----------|-------------------------------------|
| orders | → new orders placed |
| payments | → payment results (success/failure) |

inventory	→ inventory reservation results
shipping	→ shipment created/updated
order-status	→ final order status updates
notifications	→ customer notifications
*.DLT	→ dead letter for each topic

Design 3 — Real-Time Analytics Pipeline

Requirement: Clickstream analytics, dashboard updated every second

```
Web/App
  ↓
Kafka (clickstream topic)
  ↓
Kafka Streams Application
  ↓ (processes in real time)
  ├── Count clicks per page (last 1 min)
  ├── Count unique users (last 5 min)
  ├── Revenue per minute
  └── Error rate per endpoint
  ↓
Aggregated Kafka topics
  ↓           ↓
Redis (live dashboard) ClickHouse/Druid
(sub-second queries)   (historical queries)
```

Kafka Streams implementation:

```
StreamsBuilder builder = new StreamsBuilder();

KStream<String, ClickEvent> clicks =
    builder.stream("clickstream");

// Page views per minute
clicks
    .groupBy((key, click) -> click.getPageId())
    .windowedBy(
        TimeWindows.ofSizeWithNoGrace(Duration.ofMinutes(1)))
    .count()
    .toStream()
    .map((windowedKey, count) -> KeyValue.pair(
        windowedKey.key(),
        new PageViewCount(windowedKey.key(), count,
            windowedKey.window().startTime())))
    .to("page-view-counts");

// Revenue per minute
clicks
    .filter((key, click) -> click.getEventType())
```

```

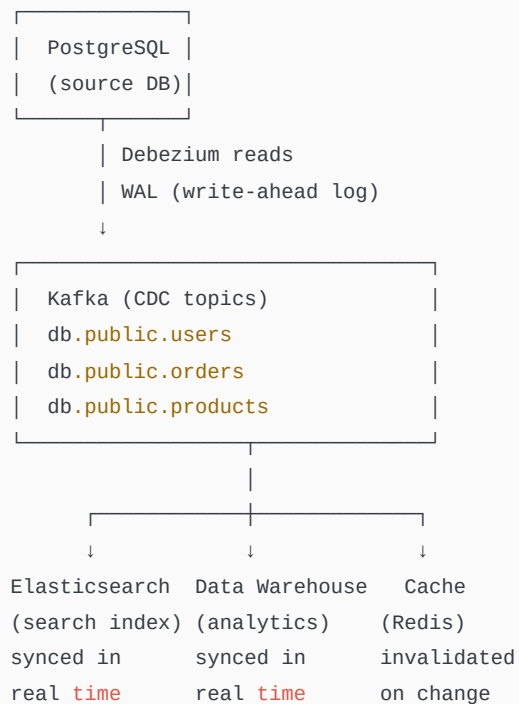
        .equals("PURCHASE"))
    .groupBy((key, click) -> click.getCategoryId())
    .windowedBy(
        TimeWindows.ofSizeWithNoGrace(Duration.ofMinutes(1)))
    .aggregate(
        () -> 0.0,
        (key, click, total) -> total + click.getAmount())
    .toStream()
    .to("revenue-per-minute");

// Unique users in 5-minute sliding window
clicks
    .groupByKey()
    .windowedBy(
        SlidingWindows.ofTimeDifferenceWithNoGrace(
            Duration.ofMinutes(5)))
    .count()
    .toStream()
    .to("unique-users");

```

Design 4 — CDC Pipeline (Change Data Capture)

Requirement: Sync database changes to multiple downstream systems in real time



Debezium message format:

```

{
  "op": "u",

```



```

"before": {
  "id": 123,
  "email": "old@email.com",
  "status": "active"
},
"after": {
  "id": 123,
  "email": "new@email.com",
  "status": "active"
},
"source": {
  "table": "users",
  "ts_ms": 1704067200000
}
}

```

Consumer handling CDC events:

```

@KafkaListener(topics = "db.public.users")
public void handleUserChange(
    ConsumerRecord<String, JsonNode> record) {

    JsonNode payload = record.value();
    String operation = payload.get("op").asText();

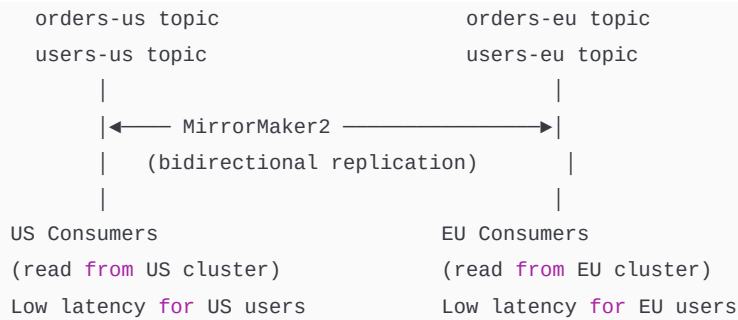
    switch (operation) {
        case "c": // INSERT
            handleInsert(payload.get("after"));
            break;
        case "u": // UPDATE
            handleUpdate(payload.get("before"),
                payload.get("after"));
            break;
        case "d": // DELETE
            handleDelete(payload.get("before"));
            break;
        case "r": // SNAPSHOT READ
            handleSnapshot(payload.get("after"));
            break;
    }
}

```

Design 5 — Multi-Region Kafka Architecture

Requirement: Global app, data in US and EU, low latency for both regions

US Region	EU Region
US Kafka Cluster	EU Kafka Cluster



Global topics (replicated both ways):

orders-us → mirrored as → us.orders-us in EU
orders-eu → mirrored as → eu.orders-eu in US

Key design considerations:

1. Topic naming convention
 - orders-us → US originated orders
 - orders-eu → EU originated orders
 - MirrorMaker prefixes: us.orders-eu in US cluster
2. Avoid replication loops
 - MirrorMaker2 tracks replication source
 - Won't re-replicate already-replicated messages
3. Consumer offset translation
 - MM2 syncs consumer offsets between clusters
 - Failover: consumers can switch cluster with correct offset
4. Data sovereignty (GDPR)
 - EU user data must stay in EU
 - Use topic-level replication rules
 - replicate.topic.filter: orders-us # only replicate US orders to EU

System Design Interview Framework for Kafka

When asked "Design X using Kafka" always cover:

1. REQUIREMENTS CLARIFICATION
 - ☐ How many events per second?
 - ☐ What latency is acceptable?
 - ☐ How long to retain data?
 - ☐ Exactly-once needed?
 - ☐ Multi-region?
2. CAPACITY ESTIMATION
 - ☐ Events/sec → partitions needed
 - ☐ Message size → storage needed
 - ☐ Retention period → total disk

- Replication factor → multiply storage

3. TOPIC DESIGN

- One topic per event type or domain?
- Partition count (parallelism)
- Retention policy
- Compaction vs time-based?
- DLT for each topic?

4. PRODUCER DESIGN

- Partition key choice (ordering)
- acks setting (reliability)
- Idempotence (duplicates)
- Compression type
- Batching strategy

5. CONSUMER DESIGN

- Consumer group per service
- How many consumers?
- Manual vs auto commit
- Error handling strategy
- Retry + DLT pattern

6. RELIABILITY

- Exactly-once or at-least-once?
- Outbox pattern for DB + Kafka?
- Schema Registry for evolution?

7. MONITORING

- Consumer lag alerts
- Under-replicated partition alerts
- Throughput dashboards

Performance vs Reliability Tradeoff Matrix

Setting	Max Performance	Max Reliability
acks	0 or 1	all
retries	0	MAX_INT
idempotence	false	true
replication	1	3
min.insync	1	2
compression	none	lz4/zstd
linger.ms	0	5-20ms
auto.commit	true	false
isolation.level	read_uncommitted	read_committed

Rule: **Start with** max reliability **in** production.
Loosen **only when** you have a measured bottleneck.

Want me to go even deeper on any specific design — like designing a fraud detection system with Kafka Streams, or a notification system, or walk through a mock system design interview end to end?

5/3/2026, 7:25:50 AM